

专题四：解三角形专题

答案版

核心公式与方法梳理

基本定理

在 $\triangle ABC$ 中，内角和与半角关系为

$$A + B + C = \pi, \quad \frac{A}{2} + \frac{B}{2} = \frac{\pi}{2} - \frac{C}{2}.$$

余弦定理及其变形为

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A, & \cos A &= \frac{b^2 + c^2 - a^2}{2bc}, \\ b^2 &= a^2 + c^2 - 2ac \cos B, & \cos B &= \frac{a^2 + c^2 - b^2}{2ac}, \\ c^2 &= a^2 + b^2 - 2ab \cos C, & \cos C &= \frac{a^2 + b^2 - c^2}{2ab}. \end{aligned}$$

正弦定理为

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R,$$

其中 R 为外接圆半径。因此

$$a = 2R \sin A, \quad b = 2R \sin B, \quad c = 2R \sin C.$$

射影定理可写为

$$a \cos B + b \cos A = c, \quad a \cos C + c \cos A = b, \quad b \cos C + c \cos B = a.$$

面积公式

记三角形面积为 S 、内切圆半径为 r 、半周长为 $s = \frac{1}{2}(a + b + c)$ ，则

$$\begin{aligned} S &= \frac{1}{2}ah_a = \frac{1}{2}bh_b = \frac{1}{2}ch_c, \\ S &= \frac{1}{2}ab \sin C = \frac{1}{2}ac \sin B = \frac{1}{2}bc \sin A, \\ S &= \frac{abc}{4R} = \sqrt{s(s-a)(s-b)(s-c)} = sr. \end{aligned}$$

从面积公式还可得到

$$\sin A = \frac{2S}{bc}, \quad \sin B = \frac{2S}{ac}, \quad \sin C = \frac{2S}{ab}.$$

三角恒等变换

基本关系为 $\sin^2 \alpha + \cos^2 \alpha = 1$ 、 $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$ 。和差角公式为

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta,$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}.$$

二倍角及降幂公式为

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha,$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha,$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha},$$

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}, \quad \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}.$$

解题提醒

先画图并标出已知量，再判断应补充哪一个边或角；含边与角的等式通常优先用正弦定理或余弦定理进行边角互化。三角函数最值问题可转化为 $y = M \sin(\omega x + \varphi) + N$ 的形式。角平分线问题可结合角平分线定理及同高三三角形的面积比；中线问题可使用

$$m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}.$$

SSA 情形中，由 $\sin B = \frac{b \sin A}{a}$ 先判断解的存在性，再结合角的范围确定是一解还是两解。判断三角形形状时，应把恒等式转化为角关系或边关系。例如 $a^2 \tan B = b^2 \tan A$ 可推出 $A = B$ 或 $C = \frac{\pi}{2}$ ，故三角形为等腰三角形或直角三角形。

题型一：边角互化与余弦定理

1. (2012 • 新课标) 已知 a, b, c 分别为 $\triangle ABC$ 三个内角 A, B, C 的对边, $a \cos C + \sqrt{3} a \sin C - b - c = 0$

(1) 求 A ;

(2) 若 $a=2$, $\triangle ABC$ 的面积为 $\sqrt{3}$; 求 b, c .

解答 【解答】解: (1) 由正弦定理得: $a \cos C + \sqrt{3} a \sin C - b - c = 0$,

即 $\sin A \cos C + \sqrt{3} \sin A \sin C = \sin B + \sin C$

$\therefore \sin A \cos C + \sqrt{3} \sin A \sin C = \sin(A+C) + \sin C$,

即 $\sqrt{3}\sin A - \cos A = 1$

$$\therefore \sin(A - 30^\circ) = \frac{1}{2}.$$

$$\therefore A - 30^\circ = 30^\circ$$

$$\therefore A = 60^\circ;$$

$$(2) \text{ 若 } a=2, \triangle ABC \text{ 的面积} = \frac{1}{2}bc \sin A = \frac{\sqrt{3}}{4}bc = \sqrt{3},$$

$$\therefore bc=4. \text{ ①}$$

再利用余弦定理可得: $a^2 = b^2 + c^2 - 2bc \cdot \cos A$

$$= (b+c)^2 - 2bc - bc = (b+c)^2 - 3 \times 4 = 4,$$

$$\therefore b+c=4. \text{ ②}$$

结合①②求得 $b=c=2$.

□

2. (2016 • 新课标 I) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $2\cos C (\cos B + \cos A) = c$.

(I) 求 C ;

(II) 若 $c = \sqrt{7}$, $\triangle ABC$ 的面积为 $\frac{3\sqrt{3}}{2}$, 求 $\triangle ABC$ 的周长.

解答 【解答】(I) 解法一 (正弦定理): $\because$ 在 $\triangle ABC$ 中, $0 < C < \pi$, $\therefore \sin C \neq 0$

已知等式利用正弦定理化简得: $2\cos C (\sin A \cos B + \sin B \cos A) = \sin C$,

整理得: $2\cos C \sin(A+B) = \sin C$,

$$\text{即 } 2\cos C \sin(\pi - (A+B)) = \sin C,$$

$$2\cos C \sin C = \sin C$$

$$\therefore \cos C = \frac{1}{2},$$

$$\therefore C = \frac{\pi}{3}.$$

解法二 (射影定理): 由 $a \cos B + b \cos A = c$, 原条件可化为

$$2c \cos C = c.$$

因为 $c > 0$, 所以 $\cos C = \frac{1}{2}$, 故 $C = \frac{\pi}{3}$.

(II) 由余弦定理得 $7 = a^2 + b^2 - 2ab \cdot \frac{1}{2}$,

$$\therefore (a+b)^2 - 3ab = 7,$$

$$\therefore S = \frac{1}{2}ab \sin C = \frac{\sqrt{3}}{4}ab = \frac{3\sqrt{3}}{2},$$

$$\therefore ab = 6,$$

$$\therefore (a+b)^2 - 18 = 7,$$

$$\therefore a+b = 5,$$

$$\therefore \triangle ABC \text{ 的周长为 } 5 + \sqrt{7}.$$

□

3. (2017 • 新课标 I) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\triangle ABC$ 的面积为 $\frac{a^2}{3 \sin A}$.

(1) 求 $\sin B \sin C$;

(2) 若 $6\cos B \cos C = 1$, $a = 3$, 求 $\triangle ABC$ 的周长.

解答 【解答】解: (1) 由三角形的面积公式可得 $S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{a^2}{3 \sin A}$,

$$\therefore 3c \sin B \sin A = 2a,$$

由正弦定理可得 $3 \sin C \sin B \sin A = 2 \sin A$,

$$\because \sin A \neq 0,$$

$$\therefore \sin B \sin C = \frac{2}{3};$$

(2) $\because 6\cos B \cos C = 1$,

$$\therefore \cos B \cos C = \frac{1}{6},$$

$$\therefore \cos B \cos C - \sin B \sin C = \frac{1}{6} - \frac{2}{3} = -\frac{1}{2},$$

$$\therefore \cos (B+C) = -\frac{1}{2},$$

$$\therefore \cos A = \frac{1}{2},$$

$$\because 0 < A < \pi,$$

$$\therefore A = \frac{\pi}{3},$$

$$\because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R = \frac{3}{\frac{\sqrt{3}}{2}} = 2\sqrt{3},$$

$$\therefore \sin B \sin C = \frac{8}{2R} \cdot \frac{c}{2R} = \frac{bc}{(2\sqrt{3})^2} = \frac{bc}{12} = \frac{2}{3},$$

$$\therefore bc = 8,$$

$$\because a^2 = b^2 + c^2 - 2bc \cos A,$$

$$\therefore b^2 + c^2 - bc = 9,$$

$$\therefore (b+c)^2 = 9 + 3cb = 9 + 24 = 33,$$

$$\therefore b+c = \sqrt{33}$$

$$\therefore \text{周长 } a+b+c = 3 + \sqrt{33}.$$

□

题型二：三角形面积

4. (2015 • 新课标 I) 已知 a, b, c 分别是 $\triangle ABC$ 内角 A, B, C 的对边, $\sin^2 B = 2 \sin A \sin C$.

(I) 若 $a = b$, 求 $\cos B$;

(II) 设 $B = 90^\circ$, 且 $a = \sqrt{2}$, 求 $\triangle ABC$ 的面积.

解答 【解答】解: (I) $\because \sin^2 B = 2 \sin A \sin C$,

由正弦定理可得: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \frac{1}{k} > 0$,

代入可得 $(bk)^2 = 2ak \cdot ck$,

$$\therefore b^2 = 2ac,$$

$$\because a = b, \therefore a = 2c,$$

$$\text{由余弦定理可得: } \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + \frac{1}{4}a^2 - a^2}{2a \times \frac{1}{2}a} = \frac{1}{4}.$$

$$\begin{aligned}
& \text{(II) 由 (I) 可得: } b^2=2ac, \\
& \therefore B=90^\circ, \text{ 且 } a=\sqrt{2}, \\
& \therefore a^2+c^2=b^2=2ac, \text{ 解得 } a=c=\sqrt{2}. \\
& \therefore S_{\triangle ABC}=\frac{1}{2}ac=1.
\end{aligned}$$

□

5. (2017 • 新课标 II) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin(A+C)=8\sin^2\frac{E}{z}$.

(1) 求 $\cos B$;

(2) 若 $a+c=6$, $\triangle ABC$ 的面积为 2, 求 b .

解答 【解答】解: (1) $\sin(A+C)=8\sin^2\frac{E}{z}$,

$$\therefore \sin B=4(1-\cos B),$$

$$\therefore \sin^2 B+\cos^2 B=1,$$

$$\therefore 16(1-\cos B)^2+\cos^2 B=1,$$

$$\therefore 16(1-\cos B)^2+\cos^2 B-1=0,$$

$$\therefore 16(\cos B-1)^2+(\cos B-1)(\cos B+1)=0,$$

$$\therefore (17\cos B-15)(\cos B-1)=0,$$

$$\therefore \cos B=\frac{15}{17};$$

$$(2) \text{ 由 (1) 可知 } \sin B=\frac{8}{17},$$

$$\therefore S_{\triangle ABC}=\frac{1}{2}ac \cdot \sin B=2,$$

$$\therefore ac=\frac{17}{2},$$

$$\therefore b^2=a^2+c^2-2ac\cos B=a^2+c^2-2\times\frac{17}{2}\times\frac{15}{17}$$

$$=a^2+c^2-15=(a+c)^2-2ac-15=36-17-15=4,$$

$$\therefore b=2.$$

□

6. (2017 • 新课标 III) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin A+\sqrt{3}\cos A=0$, $a=2\sqrt{7}$, $b=2$.

(1) 求 c ;

(2) 设 D 为 BC 边上一点, 且 $AD\perp AC$, 求 $\triangle ABD$ 的面积.

解答 【解答】解: (1) $\therefore \sin A+\sqrt{3}\cos A=0$,

$$\therefore \tan A=-\sqrt{3},$$

$$\therefore 0 < A < \pi,$$

$$\therefore A=\frac{2\pi}{3},$$

$$\text{由余弦定理可得 } a^2=b^2+c^2-2bccosA,$$

$$\text{即 } 28=4+c^2-2\times 2c\times (-\frac{1}{2}),$$

$$\text{即 } c^2+2c-24=0,$$

$$\text{解得 } c=-6 \text{ (舍去) 或 } c=4,$$

故 $c=4$.

$$(2) \because c^2 = b^2 + a^2 - 2ab\cos C,$$

$$\therefore 16 = 28 + 4 - 2 \times 2\sqrt{7} \times 2 \times \cos C,$$

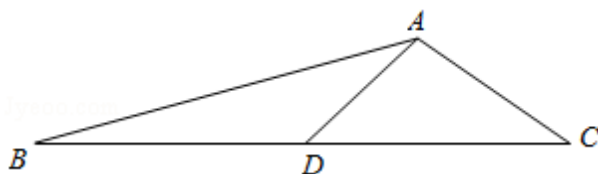
$$\therefore \cos C = \frac{2}{\sqrt{7}},$$

$$\therefore CD = \frac{AC}{\cos C} = \frac{2}{\frac{2}{\sqrt{7}}} = \sqrt{7}$$

$$\therefore CD = \frac{1}{2}BC$$

$$\therefore S_{\triangle ABC} = \frac{1}{2}AB \cdot AC \cdot \sin \angle BAC = \frac{1}{2} \times 4 \times 2 \times \frac{\sqrt{3}}{2} = 2\sqrt{3},$$

$$\therefore S_{\triangle ABD} = \frac{1}{2}S_{\triangle ABC} = \sqrt{3}$$



□

题型三：利用基本不等式求最值

7. (2013 • 新课标 II) $\triangle ABC$ 在内角 A 、 B 、 C 的对边分别为 a 、 b 、 c ，已知 $a = b\cos C + c\sin B$.

(I) 求 B ;

(II) 若 $b=2$ ，求 $\triangle ABC$ 面积的最大值.

解答 【解答】解：(I) 由已知及正弦定理得： $\sin A = \sin B\cos C + \sin B\sin C$ ①，

$$\therefore \sin A = \sin(B+C) = \sin B\cos C + \cos B\sin C$$
 ②，

$$\therefore \sin B = \cos B, \text{ 即 } \tan B = 1,$$

$\therefore B$ 为三角形的内角，

$$\therefore B = \frac{\pi}{4};$$

$$(II) S_{\triangle ABC} = \frac{1}{2}ac\sin B = \frac{\sqrt{2}}{4}ac,$$

$$\text{由已知及余弦定理得： } 4 = a^2 + c^2 - 2ac\cos \frac{\pi}{4} \geq 2ac - 2ac \times \frac{\sqrt{2}}{2},$$

$$\text{整理得： } ac \leq \frac{4}{2-\sqrt{2}}, \text{ 当且仅当 } a=c \text{ 时，等号成立，}$$

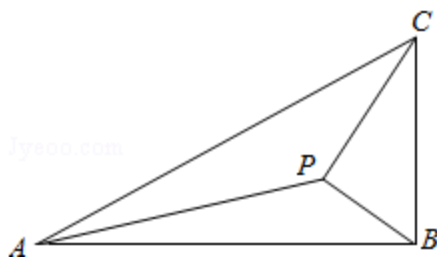
$$\text{则 } \triangle ABC \text{ 面积的最大值为 } \frac{1}{2} \times \frac{\sqrt{2}}{2} \times \frac{4}{2-\sqrt{2}} = \frac{1}{2} \times \sqrt{2} \times (2+\sqrt{2}) = \sqrt{2}+1.$$

□

题型四：图形分析与未知量表示

8. (2013 • 新课标 I) 如图，在 $\triangle ABC$ 中， $\angle ABC = 90^\circ$ ， $AB = \sqrt{3}$ ， $BC = 1$ ， P 为 $\triangle ABC$ 内一点， $\angle BPC = 90^\circ$.

- (1) 若 $PB = \frac{1}{2}$, 求 PA ;
 (2) 若 $\angle APB = 150^\circ$, 求 $\tan \angle PBA$.



解答 【解答】解： (I) 在 $\text{Rt}\triangle PBC$ 中, $\cos \angle PBC = \frac{PB}{BC} = \frac{1}{2}$, $\therefore \angle PBC = 60^\circ$,
 $\therefore \angle PBA = 30^\circ$.

在 $\triangle PBA$ 中, 由余弦定理得

$$A^2 = PB^2 + AB^2 - 2PB \cdot AB \cos 30^\circ = \left(\frac{1}{2}\right)^2 + (\sqrt{3})^2 - 2 \cdot \frac{1}{2} \cdot \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{7}{4}.$$

$$\therefore PA = \frac{\sqrt{7}}{2}.$$

(II) 设 $\angle PBA = \alpha$, 在 $\text{Rt}\triangle PBC$ 中, $PB = BC \cos (90^\circ - \alpha) = \sin \alpha$.

在 $\triangle PBA$ 中, 由正弦定理得 $\frac{AB}{\sin \angle APB} = \frac{PB}{\sin \angle PAB}$, 即 $\frac{\sqrt{3}}{\sin 150^\circ} = \frac{\sin \alpha}{\sin (30^\circ - \alpha)}$,

$$\text{化为 } \sqrt{3} \cos \alpha = 4 \sin \alpha. \therefore \tan \alpha = \frac{\sqrt{3}}{4}.$$

□

9. (2014 • 新课标 II) 四边形 $ABCD$ 的内角 A 与 C 互补, $AB=1, BC=3, CD=DA=2$.

- (1) 求 C 和 BD ;
 (2) 求四边形 $ABCD$ 的面积.

解答 【解答】解： (1) 在 $\triangle BCD$ 中, $BC=3, CD=2$,

$$\text{由余弦定理得: } BD^2 = BC^2 + CD^2 - 2BC \cdot CD \cos C = 13 - 12 \cos C \quad \text{①},$$

在 $\triangle ABD$ 中, $AB=1, DA=2, A+C=\pi$,

$$\text{由余弦定理得: } BD^2 = AB^2 + AD^2 - 2AB \cdot AD \cos A = 5 - 4 \cos A = 5 + 4 \cos C \quad \text{②},$$

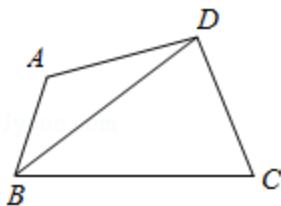
$$\text{由①②得: } \cos C = \frac{1}{2},$$

$$\text{则 } C = 60^\circ, BD = \sqrt{7};$$

$$(2) \because \cos C = \frac{1}{2}, \cos A = -\frac{1}{2},$$

$$\therefore \sin C = \sin A = \frac{\sqrt{3}}{2},$$

$$\text{则 } S = \frac{1}{2} AB \cdot DA \sin A + \frac{1}{2} BC \cdot CD \sin C = \frac{1}{2} \times 1 \times 2 \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times 3 \times 2 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}.$$



□

10. (2015 • 新课标 II) $\triangle ABC$ 中, D 是 BC 上的点, AD 平分 $\angle BAC$, $\triangle ABD$ 面积是 $\triangle ADC$ 面积的 2 倍.

(1) 求 $\frac{\sin B}{\sin C}$;

(2) 若 $AD=1$, $DC=\frac{\sqrt{2}}{2}$, 求 BD 和 AC 的长.

解答 【解答】解: (1) 如图, 过 A 作 $AE \perp BC$ 于 E,

$$\therefore \frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{\frac{1}{2}BD \cdot AE}{\frac{1}{2}DC \cdot AE} = 2$$

$$\therefore BD = 2DC,$$

$\therefore AD$ 平分 $\angle BAC$

$$\therefore \angle BAD = \angle DAC$$

$$\text{在 } \triangle ABD \text{ 中, } \frac{BD}{\sin \angle BAD} = \frac{AD}{\sin B}, \therefore \sin B = \frac{AD \sin \angle BAD}{BD}$$

$$\text{在 } \triangle ADC \text{ 中, } \frac{DC}{\sin \angle DAC} = \frac{AD}{\sin C}, \therefore \sin C = \frac{AD \sin \angle DAC}{DC};$$

$$\therefore \frac{\sin B}{\sin C} = \frac{DC}{BD} = \frac{1}{2}. \quad \cdots 6 \text{ 分}$$

(2) 由 (1) 知, $BD = 2DC = 2 \times \frac{\sqrt{2}}{2} = \sqrt{2}$.

过 D 作 $DM \perp AB$ 于 M, 作 $DN \perp AC$ 于 N,

$\therefore AD$ 平分 $\angle BAC$,

$$\therefore DM = DN,$$

$$\therefore \frac{S_{\triangle ABD}}{S_{\triangle ADC}} = \frac{\frac{1}{2}AB \cdot DM}{\frac{1}{2}AC \cdot DN} = 2,$$

$$\therefore AB = 2AC,$$

令 $AC = x$, 则 $AB = 2x$,

$\therefore \angle BAD = \angle DAC$,

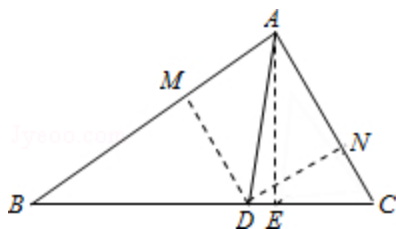
$$\therefore \cos \angle BAD = \cos \angle DAC,$$

$$\therefore \text{由余弦定理可得: } \frac{(2x)^2 + 1^2 - (\sqrt{2})^2}{2 \times 2x \times 1} = \frac{x^2 + 1^2 - (\frac{\sqrt{2}}{2})^2}{2 \times x \times 1},$$

$$\therefore x = 1,$$

$$\therefore AC = 1,$$

$\therefore BD$ 的长为 $\sqrt{2}$, AC 的长为 1.



□

11. (2018 • 新课标 I) 在平面四边形 ABCD 中, $\angle ADC = 90^\circ$, $\angle A = 45^\circ$, $AB = 2$, $BD = 5$.

(1) 求 $\cos \angle ADB$;

(2) 若 $DC = 2\sqrt{2}$, 求 BC.

解答 【解答】解: (1) $\therefore \angle ADC = 90^\circ$, $\angle A = 45^\circ$, $AB = 2$, $BD = 5$.

∴ 由正弦定理得: $\frac{AB}{\sin \angle ADB} = \frac{BD}{\sin A}$, 即 $\frac{2}{\sin \angle ADB} = \frac{5}{\sin 45^\circ}$,

$$\therefore \sin \angle ADB = \frac{2 \sin 45^\circ}{5} = \frac{\sqrt{2}}{5},$$

∵ $AB < BD$, ∴ $\angle ADB < \angle A$,

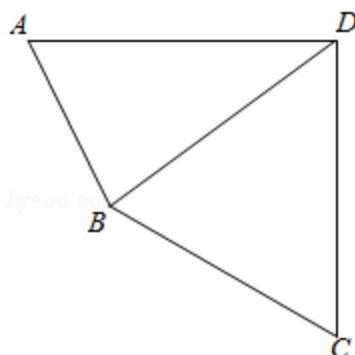
$$\therefore \cos \angle ADB = \sqrt{1 - \left(\frac{\sqrt{2}}{5}\right)^2} = \frac{\sqrt{23}}{5}.$$

(2) ∵ $\angle ADC = 90^\circ$, ∴ $\cos \angle BDC = \sin \angle ADB = \frac{\sqrt{2}}{5}$,

$$\therefore DC = 2\sqrt{2},$$

$$\therefore BC = \sqrt{BD^2 + DC^2 - 2BD \cdot DC \cos \angle BDC}$$

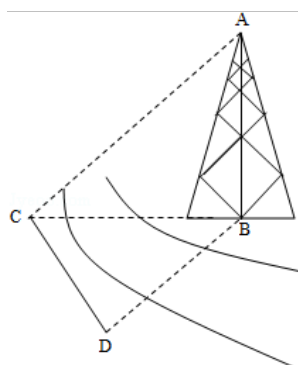
$$= \sqrt{25 + 8 - 2 \times 5 \times 2\sqrt{2} \times \frac{\sqrt{2}}{5}} = 5.$$



□

题型五：解三角形的实际应用

12. (2007 年海南省高考数学试卷(理)) 如图, 测量河对岸的塔高 AB 时, 可以选与塔底 B 在同一水平面内的两个测点 C 与 D . 现测得 $\angle BCD = \alpha$, $\angle BDC = \beta$, $CD = s$, 并在点 C 测得塔顶 A 的仰角为 θ , 求塔高 AB .



解答 【解答】解: 在 $\triangle BCD$ 中, $\angle CBD = \pi - \alpha - \beta$.

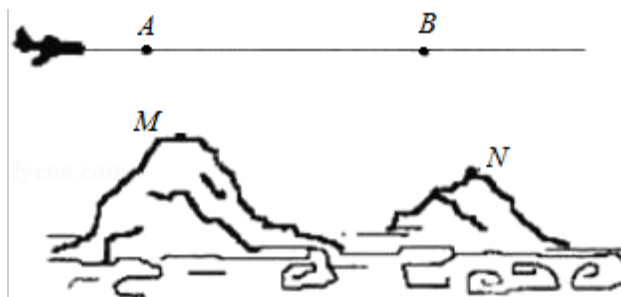
由正弦定理得 $\frac{BC}{\sin \angle BDC} = \frac{CD}{\sin \angle CBD}$.

$$\text{所以 } BC = \frac{CD \sin \angle BDC}{\sin \angle CBD} = \frac{s \sin \beta}{\sin(\alpha + \beta)}.$$

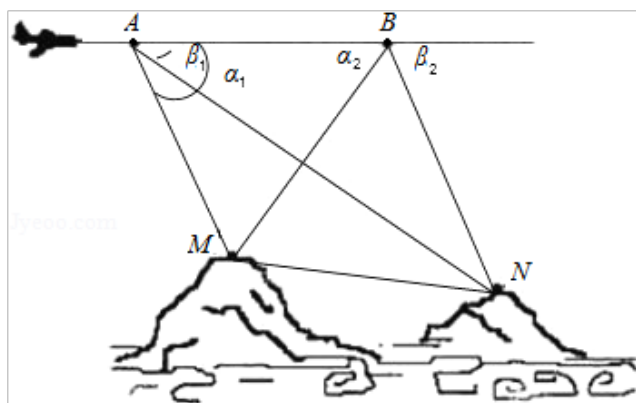
在 $\text{Rt} \triangle ABC$ 中, $AB = BC \tan \angle ACB = \frac{s \tan \theta \sin \beta}{\sin(\alpha + \beta)}.$

□

13. (12 分) (2009 • 宁夏) 为了测量两山顶 M, N 间的距离, 飞机沿水平方向在 A, B 两点进行测量, A, B, M, N 在同一个铅垂平面内 (如示意图), 飞机能够测量的数据有俯角和 A, B 间的距离, 请设计一个方案, 包括: ①指出需要测量的数据 (用字母表示, 并在图中标出); ②用文字和公式写出计算 M, N 间的距离的步骤.



解答 【解答】 方案一: ①需要测量的数据有: A 点到 M, N 点的俯角 α_1, β_1 ; B 点到 M, N 点的俯角 α_2, β_2 ; A, B 的距离 d , 如图所示:



②第一步: 计算 AM . 由正弦定理 $AM = \frac{d \sin \alpha_2}{\sin(\alpha_1 + \alpha_2)}$

第二步: 计算 AN . 由正弦定理 $AN = \frac{d \sin \beta_2}{\sin(\beta_2 - \beta_1)}$

第三步: 计算 MN . 由余弦定理 $MN = \sqrt{AM^2 + AN^2 - 2AM \cdot AN \cos(\alpha_1 - \beta_1)}$

方案二: ①需要测量的数据有: A 点到 M, N 点的俯角 α_1, β_1 ; B 点到 M, N 点的俯角 α_2, β_2 ; A, B 的距离 d (如图所示).

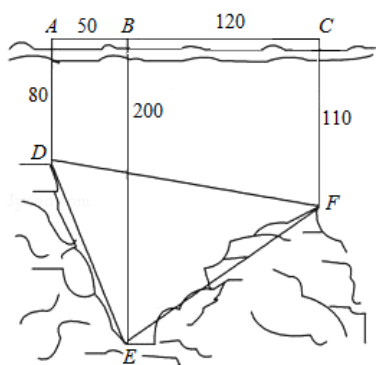
②第一步: 计算 BM . 由正弦定理 $BM = \frac{d \sin \alpha_1}{\sin(\alpha_1 + \alpha_2)}$

第二步: 计算 BN . 由正弦定理 $BN = \frac{d \sin \beta_1}{\sin(\beta_2 - \beta_1)}$

第三步: 计算 MN . 由余弦定理 $MN = \sqrt{BM^2 + BN^2 + 2BM \cdot BN \cos(\alpha_2 + \beta_2)}$

□

14. (2009 • 宁夏) 如图, 为了解某海域海底构造, 在海平面内一条直线上的 A, B, C 三点进行测量. 已知 $AB=50$ m, $BC=120$ m, 于 A 处测得水深 $AD=80$ m, 于 B 处测得水深 $BE=200$ m, 于 C 处测得水深 $CF=110$ m, 求 $\angle DEF$ 的余弦值.



解答 【解答】解：如图作 $DM \parallel AC$ 交 BE 于 N ，交 CF 于 M 。

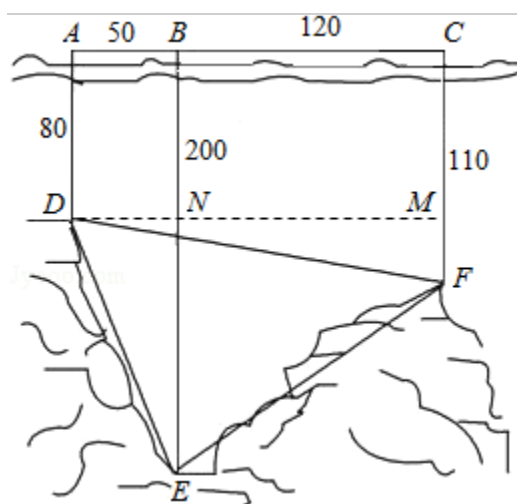
$$DF = \sqrt{MF^2 + DM^2} = \sqrt{30^2 + 170^2} = 10\sqrt{298} \text{ (m)},$$

$$DE = \sqrt{DN^2 + EN^2} = \sqrt{50^2 + 120^2} = 130 \text{ (m)},$$

$$EF = \sqrt{(BE - FC)^2 + BC^2} = \sqrt{90^2 + 120^2} = 150 \text{ (m)}.$$

在 $\triangle DEF$ 中，由余弦定理的变形公式，得

$$\cos \angle DEF = \frac{DE^2 + EF^2 - DF^2}{2DE \cdot EF} = \frac{130^2 + 150^2 - 10^2 \cdot 298}{2 \cdot 130 \cdot 150} = \frac{16}{65}.$$



□

题型六：常用公式综合

15. (2017 • 天津) 在 $\triangle ABC$ 中，内角 A, B, C 所对的边分别为 a, b, c . 已知 $a \sin A = 4b \sin B$, $ac = \sqrt{5}(a^2 - b^2 - c^2)$.

(I) 求 $\cos A$ 的值;

(II) 求 $\sin(2B - A)$ 的值.

解答 【解答】(I) 解：由 $\frac{a}{\sin A} = \frac{b}{\sin B}$ ，得 $a \sin B = b \sin A$ ，

又 $a \sin A = 4b \sin B$ ，得 $4b \sin B = a \sin A$ ，

两式作比得： $\frac{a}{4a} = \frac{b}{a}$, $\therefore a = 2b$.

由 $ac = \sqrt{5}(a^2 - b^2 - c^2)$ ，得 $b^2 + c^2 - a^2 = \frac{\sqrt{5}}{5}ac$,

由余弦定理, 得 $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{-\frac{\sqrt{5}}{5}ac}{ac} = -\frac{\sqrt{5}}{5}$;

(II) 解: 由 (I), 可得 $\sin A = \frac{2\sqrt{5}}{5}$, 代入 $a\sin A = 4b\sin B$, 得 $\sin B = \frac{a\sin A}{4b} = \frac{\sqrt{5}}{5}$.

由 (I) 知, A 为钝角, 则 B 为锐角,

$$\therefore \cos B = \sqrt{1 - \sin^2 B} = \frac{2\sqrt{5}}{5}.$$

$$\text{于是 } \sin 2B = 2 \sin B \cos B = \frac{4}{5}, \quad \cos 2B = 1 - 2 \sin^2 B = \frac{3}{5},$$

$$\text{故 } \sin(2B - A) = \sin 2B \cos A - \cos 2B \sin A = -\frac{2\sqrt{5}}{5}.$$

□

题型七：三角形内角和

16. (2016 • 浙江) 在 $\triangle ABC$ 中, 内角 A, B, C 所对的边分别为 a, b, c, 已知 $b+c=2a\cos B$.

(I) 证明: $A=2B$;

(II) 若 $\triangle ABC$ 的面积 $S = \frac{a^2}{4}$, 求角 A 的大小.

解答 【解答】(I) 证明: $\because b+c=2a\cos B$,

$$\therefore \sin B + \sin C = 2\sin A \cos B,$$

$$\therefore \sin B + \sin(A+B) = 2\sin A \cos B$$

$$\therefore \sin B + \sin A \cos B + \cos A \sin B = 2\sin A \cos B$$

$$\therefore \sin B = \sin A \cos B - \cos A \sin B = \sin(A-B)$$

$\because A, B$ 是三角形中的角,

$$\therefore B = A-B,$$

$$\therefore A = 2B;$$

(II) 解: $\because \triangle ABC$ 的面积 $S = \frac{a^2}{4}$,

$$\therefore \frac{1}{2}bc\sin A = \frac{a^2}{4},$$

$$\therefore 2bc\sin A = a^2,$$

$$\therefore 2\sin B \sin C = \sin A = \sin 2B,$$

$$\therefore \sin C = \cos B,$$

$$\therefore B+C=90^\circ, \text{ 或 } C=B+90^\circ,$$

$$\therefore A=90^\circ \text{ 或 } A=45^\circ.$$

□

题型八：三角函数化简与最值

17. (2016 • 北京) 在 $\triangle ABC$ 中, $a^2+c^2=b^2+\sqrt{2}ac$.

(I) 求 $\angle B$ 的大小;

(II) 求 $\sqrt{2}\cos A + \cos C$ 的最大值.

解答 【解答】解: (I) $\because$ 在 $\triangle ABC$ 中, $a^2+c^2=b^2+\sqrt{2}ac$.

$$\therefore a^2 + c^2 - b^2 = \sqrt{2}ac.$$

$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{\sqrt{2}ac}{2ac} = \frac{\sqrt{2}}{2},$$

$$\therefore B = \frac{\pi}{4}$$

(II) 由 (I) 得: $C = \frac{3\pi}{4} - A$,

$$\therefore \sqrt{2}\cos A + \cos C = \sqrt{2}\cos A + \cos\left(\frac{3\pi}{4} - A\right)$$

$$= \sqrt{2}\cos A - \frac{\sqrt{2}}{2}\cos A + \frac{\sqrt{2}}{2}\sin A$$

$$= \frac{\sqrt{2}}{2}\cos A + \frac{\sqrt{2}}{2}\sin A$$

$$= \sin\left(A + \frac{\pi}{4}\right).$$

$$\therefore A \in \left(0, \frac{3\pi}{4}\right),$$

$$\therefore A + \frac{\pi}{4} \in \left(\frac{\pi}{4}, \pi\right),$$

故当 $A + \frac{\pi}{4} = \frac{\pi}{2}$ 时, $\sin\left(A + \frac{\pi}{4}\right)$ 取最大值 1,

即 $\sqrt{2}\cos A + \cos C$ 的最大值为 1.

□

题型九：解三角形与数列综合

18. (2014 • 陕西) $\triangle ABC$ 的内角 A, B, C 所对应的边分别为 a, b, c .

(I) 若 a, b, c 成等差数列, 证明: $\sin A + \sin C = 2\sin(A + C)$;

(II) 若 a, b, c 成等比数列, 求 $\cos B$ 的最小值.

解答 【解答】解: (I) $\because a, b, c$ 成等差数列,

$$\therefore 2b = a + c,$$

利用正弦定理化简得: $2\sin B = \sin A + \sin C$,

$$\therefore \sin B = \sin[\pi - (A + C)] = \sin(A + C),$$

$$\therefore \sin A + \sin C = 2\sin B = 2\sin(A + C);$$

(II) $\because a, b, c$ 成等比数列,

$$\therefore b^2 = ac,$$

$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + c^2 - ac}{2ac} \geq \frac{2ac - ac}{2ac} = \frac{1}{2},$$

当且仅当 $a = c$ 时等号成立,

$$\therefore \cos B \text{ 的最小值为 } \frac{1}{2}.$$

□

历年高考解三角形大题

19. (2018 • 北京) 在 $\triangle ABC$ 中, $a = 7, b = 8, \cos B = -\frac{1}{7}$.

(I) 求 $\angle A$;

(II) 求 AC 边上的高.

解答 【解答】解: (I) $\because a < b, \therefore A < B$, 即 A 是锐角,

$$\because \cos B = -\frac{1}{7}, \therefore \sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \left(-\frac{1}{7}\right)^2} = \frac{4\sqrt{3}}{7},$$

$$\text{由正弦定理得 } \frac{a}{\sin A} = \frac{b}{\sin B} \text{ 得 } \sin A = \frac{a \sin B}{b} = \frac{7 \times \frac{4\sqrt{3}}{7}}{8} = \frac{\sqrt{3}}{2},$$

$$\text{则 } A = \frac{\pi}{3}.$$

$$\text{(II) 由余弦定理得 } b^2 = a^2 + c^2 - 2ac \cos B,$$

$$\text{即 } 64 = 49 + c^2 + 2 \times 7 \times c \times \frac{1}{7},$$

$$\text{即 } c^2 + 2c - 15 = 0,$$

$$\text{得 } (c-3)(c+5) = 0,$$

$$\text{得 } c = 3 \text{ 或 } c = -5 \text{ (舍)},$$

$$\text{则 AC 边上的高 } h = c \sin A = 3 \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}.$$

□

20. (2018 • 天津) 在 $\triangle ABC$ 中, 内角 A, B, C 所对的边分别为 a, b, c . 已知 $b \sin A = a \cos \left(B - \frac{\pi}{6}\right)$.

(I) 求角 B 的大小;

(II) 设 $a = 2, c = 3$, 求 b 和 $\sin(2A - B)$ 的值.

解答 【解答】解: (I) 在 $\triangle ABC$ 中, 由正弦定理得 $\frac{a}{\sin A} = \frac{b}{\sin B}$, 得 $b \sin A = a \sin B$,

$$\text{又 } b \sin A = a \cos \left(B - \frac{\pi}{6}\right).$$

$$\therefore a \sin B = a \cos \left(B - \frac{\pi}{6}\right), \text{ 即 } \sin B = \cos \left(B - \frac{\pi}{6}\right) = \cos B \cos \frac{\pi}{6} + \sin B \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} \cos B + \frac{1}{2} \sin B,$$

$$\therefore \tan B = \sqrt{3},$$

$$\text{又 } B \in (0, \pi), \therefore B = \frac{\pi}{3}.$$

$$\text{(II) 在 } \triangle ABC \text{ 中, } a = 2, c = 3, B = \frac{\pi}{3},$$

$$\text{由余弦定理得 } b = \sqrt{a^2 + c^2 - 2ac \cos B} = \sqrt{7}, \text{ 由 } b \sin A = a \cos \left(B - \frac{\pi}{6}\right), \text{ 得 } \sin A = \frac{\sqrt{3}}{\sqrt{7}},$$

$$\because a < c, \therefore \cos A = \frac{2}{\sqrt{7}},$$

$$\therefore \sin 2A = 2 \sin A \cos A = \frac{4\sqrt{3}}{7},$$

$$\cos 2A = 2 \cos^2 A - 1 = \frac{1}{7},$$

$$\therefore \sin(2A - B) = \sin 2A \cos B - \cos 2A \sin B = \frac{4\sqrt{3}}{7} \cdot \frac{1}{2} - \frac{1}{7} \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{14}.$$

□

21. (2018 • 全国) 在 $\triangle ABC$ 中, 角 A, B, C 对应边 a, b, c , 外接圆半径为 1, 已知 $2(\sin^2 A - \sin^2 C) = (a - b) \sin B$.

(1) 证明 $a^2 + b^2 - c^2 = ab$;

(2) 求角 C 和边 c .

解答 【解答】证明：(1) $\because$ 在 $\triangle ABC$ 中，角 A 、 B 、 C 对应边 a 、 b 、 c ，外接圆半径为 1，

$$\therefore \text{由正弦定理得：} \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R = 2,$$

$$\therefore \sin A = \frac{a}{2}, \sin B = \frac{b}{2}, \sin C = \frac{c}{2},$$

$$\therefore 2(\sin^2 A - \sin^2 C) = (a-b) \sin B,$$

$$\therefore 2\left(\frac{a^2}{4} - \frac{c^2}{4}\right) = (a-b) \cdot \frac{b}{2},$$

$$\text{化简，得：} a^2 + b^2 - c^2 = ab,$$

$$\text{故 } a^2 + b^2 - c^2 = ab.$$

$$\text{解：(2) } \because a^2 + b^2 - c^2 = ab,$$

$$\therefore \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{ab}{2ab} = \frac{1}{2},$$

$$\text{解得 } C = \frac{\pi}{3},$$

$$\therefore c = 2\sin C = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}.$$

□

22. (2019 • 新课标 I) $\triangle ABC$ 的内角 A 、 B 、 C 的对边分别为 a 、 b 、 c 。设 $(\sin B - \sin C)^2 = \sin^2 A - \sin B \sin C$ 。

(1) 求 A ；

(2) 若 $\sqrt{2}a + b = 2c$ ，求 $\sin C$ 。

解答 【解答】解：(1) $\triangle ABC$ 的内角 A 、 B 、 C 的对边分别为 a 、 b 、 c 。

$$\because (\sin B - \sin C)^2 = \sin^2 A - \sin B \sin C,$$

$$\therefore \sin^2 B + \sin^2 C - 2\sin B \sin C = \sin^2 A - \sin B \sin C,$$

$$\therefore \text{由正弦定理得：} b^2 + c^2 - a^2 = bc,$$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{bc}{2bc} = \frac{1}{2},$$

$$\because 0 < A < \pi, \therefore A = \frac{\pi}{3}.$$

$$(2) \because \sqrt{2}a + b = 2c, A = \frac{\pi}{3},$$

$$\therefore \text{由正弦定理得 } \sqrt{2}\sin A + \sin B = 2\sin C,$$

$$\therefore \frac{\sqrt{6}}{2} + \sin\left(\frac{2\pi}{3} - C\right) = 2\sin C$$

$$\text{解得 } \sin\left(C - \frac{\pi}{6}\right) = \frac{\sqrt{2}}{2},$$

$$\therefore C - \frac{\pi}{6} = \frac{\pi}{4} \text{ 或 } C - \frac{\pi}{6} = \frac{3\pi}{4},$$

$$\because 0 < C < \frac{2\pi}{3}, \therefore C = \frac{\pi}{4} + \frac{\pi}{6},$$

$$\therefore \sin C = \sin\left(\frac{\pi}{4} + \frac{\pi}{6}\right) = \sin\frac{\pi}{4}\cos\frac{\pi}{6} + \cos\frac{\pi}{4}\sin\frac{\pi}{6} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

□

23. (2019 • 北京) 在 $\triangle ABC$ 中， $a = 3$ ， $b - c = 2$ ， $\cos B = -\frac{1}{2}$ 。

(I) 求 b 、 c 的值；

(II) 求 $\sin(B + C)$ 的值。

(III) 求 $\sin(B - C)$ 的值。

解答 【解答】解：(1) $\because a = 3, b - c = 2, \cos B = -\frac{1}{2}$.

$\therefore$ 由余弦定理，得 $b^2 = a^2 + c^2 - 2ac \cos B$

$$= 9 + (b-2)^2 - 2 \times 3 \times (b-2) \times \left(-\frac{1}{2}\right),$$

$$\therefore b = 7, \therefore c = b - 2 = 5;$$

(2) 在 $\triangle ABC$ 中， $\because \cos B = -\frac{1}{2}, \therefore \sin B = \frac{\sqrt{3}}{2}$,

由正弦定理有： $\frac{a}{\sin A} = \frac{b}{\sin B}$,

$$\therefore \sin A = \frac{a \sin B}{b} = \frac{3 \cdot \frac{\sqrt{3}}{2}}{7} = \frac{3\sqrt{3}}{14},$$

$$\therefore \sin(B+C) = \sin(\pi - A) = \sin A = \frac{3\sqrt{3}}{14}.$$

(III) 由正弦定理，

$$\sin C = \frac{c \sin B}{b} = \frac{5\sqrt{3}}{14}.$$

因为 $b > c$ ，所以 $B > C$ ，从而 C 为锐角，

$$\cos C = \sqrt{1 - \sin^2 C} = \frac{11}{14}.$$

因此

$$\sin(B-C) = \sin B \cos C - \cos B \sin C = \frac{4\sqrt{3}}{7}.$$

□

24. (2019 • 江苏) 在 $\triangle ABC$ 中，角 A, B, C 的对边分别为 a, b, c .

(1) 若 $a = 3c, b = \sqrt{2}, \cos B = \frac{2}{3}$ ，求 c 的值；

(2) 若 $\frac{\sin A}{a} = \frac{\cos B}{2b}$ ，求 $\sin(B + \frac{\pi}{2})$ 的值.

解答 【解答】解：(1) $\because$ 在 $\triangle ABC$ 中，角 A, B, C 的对边分别为 a, b, c .

$$a = 3c, b = \sqrt{2}, \cos B = \frac{2}{3},$$

$\therefore$ 由余弦定理得：

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{10c^2 - 2}{6c^2} = \frac{2}{3},$$

$$\text{解得 } c = \frac{\sqrt{3}}{3}.$$

$$(2) \because \frac{\sin A}{a} = \frac{\cos B}{2b},$$

$$\therefore \text{由正弦定理得：} \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\cos B}{2b},$$

$$\therefore 2\sin B = \cos B,$$

$$\therefore \sin^2 B + \cos^2 B = 1,$$

$$\therefore \sin B = \frac{\sqrt{5}}{5}, \cos B = \frac{2\sqrt{5}}{5},$$

$$\therefore \sin(B + \frac{\pi}{2}) = \cos B = \frac{2\sqrt{5}}{5}.$$

□

25. (2019 • 天津) 在 $\triangle ABC$ 中, 内角 A, B, C 所对的边分别为 a, b, c . 已知 $b+c=2a$, $3c\sin B=4a\sin C$.

(I) 求 $\cos B$ 的值;

(II) 求 $\sin(2B+\frac{\pi}{6})$ 的值.

解答 【解答】解 (I) 在三角形 ABC 中, 由正弦定理 $\frac{b}{\sin B}=\frac{c}{\sin C}$, 得 $b\sin C=c\sin B$, 又由 $3c\sin B=4a\sin C$,

得 $3b\sin C=4a\sin C$, 即 $3b=4a$. 又因为 $b+c=2a$, 得 $b=\frac{4a}{3}$, $c=\frac{2a}{3}$, 由

余弦定理可得 $\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{a^2+\frac{4}{9}a^2-\frac{16}{9}a^2}{2a\cdot\frac{2}{3}a}=-\frac{1}{4}$.

(II) 由 (I) 得 $\sin B=\sqrt{1-\cos^2 B}=\frac{\sqrt{15}}{4}$, 从而 $\sin 2B=2\sin B\cos B=-\frac{\sqrt{15}}{8}$,
 $\cos 2B=\cos^2 B-\sin^2 B=-\frac{7}{8}$,

故 $\sin(2B+\frac{\pi}{6})=\sin 2B\cos\frac{\pi}{6}+\cos 2B\sin\frac{\pi}{6}=-\frac{\sqrt{15}}{8}\times\frac{\sqrt{3}}{2}-\frac{7}{8}\times\frac{1}{2}=-\frac{3\sqrt{5}+7}{16}$.

□

26. (2019 • 新课标III) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c . 已知 $a\sin\frac{\pi}{2}-C=b\sin A$.

(1) 求 B ;

(2) 若 $\triangle ABC$ 为锐角三角形, 且 $c=1$, 求 $\triangle ABC$ 面积的取值范围.

解答 【解答】解: (1) $a\sin\frac{\pi}{2}-C=b\sin A$, 即为 $a\sin\frac{7-\pi}{2}=a\cos\frac{E}{2}=b\sin A$,

可得 $\sin A\cos\frac{E}{2}=\sin B\sin A=2\sin\frac{E}{2}\cos\frac{E}{2}\sin A$,

$\therefore \sin A>0$,

$\therefore \cos\frac{E}{2}=2\sin\frac{E}{2}\cos\frac{E}{2}$,

若 $\cos\frac{E}{2}=0$, 可得 $B=(2k+1)\pi$, $k\in\mathbb{Z}$ 不成立,

$\therefore \sin\frac{E}{2}=\frac{1}{2}$,

由 $0<B<\pi$, 可得 $B=\frac{\pi}{3}$;

(2) 若 $\triangle ABC$ 为锐角三角形, 且 $c=1$,

由余弦定理可得 $b=\sqrt{a^2+1-2a\cos\frac{\pi}{3}}=\sqrt{a^2-a+1}$,

由三角形 ABC 为锐角三角形, 可得 $a^2+a^2-a+1>1$ 且 $1+a^2-a+1>a^2$, 且 $1+a^2>a^2-a+1$,

解得 $\frac{1}{2}<a<2$,

可得 $\triangle ABC$ 面积 $S=\frac{1}{2}a\cdot\sin\frac{\pi}{3}=\frac{\sqrt{3}}{4}a\in(\frac{\sqrt{3}}{8}, \frac{\sqrt{3}}{2})$.

□

27. (2020 • 新课标II) $\triangle ABC$ 中, $\sin^2 A-\sin^2 B-\sin^2 C=\sin B\sin C$.

(1) 求 A ;

(2) 若 $BC=3$, 求 $\triangle ABC$ 周长的最大值.

解答 【解答】解: (1) 设 $\triangle ABC$ 的内角 A, B, C 所对的边分别为 a, b, c ,

因为 $\sin^2 A - \sin^2 B - \sin^2 C = \sin B \sin C$,

由正弦定理可得 $a^2 - b^2 - c^2 = bc$,

即为 $b^2 + c^2 - a^2 = -bc$,

由余弦定理可得 $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = -\frac{bc}{2bc} = -\frac{1}{2}$,

由 $0 < A < \pi$, 可得 $A = \frac{2\pi}{3}$;

(2) 由题意可得 $a = 3$,

又 $B + C = \frac{\pi}{3}$, 可设 $B = \frac{\pi}{6} - d$, $C = \frac{\pi}{6} + d$, $-\frac{\pi}{6} < d < \frac{\pi}{6}$,

由正弦定理可得 $\frac{3}{\sin \frac{2\pi}{3}} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2\sqrt{3}$,

可得 $b = 2\sqrt{3}\sin(\frac{\pi}{6} - d)$, $c = 2\sqrt{3}\sin(\frac{\pi}{6} + d)$,

则 $\triangle ABC$ 周长为 $a + b + c = 3 + 2\sqrt{3}[\sin(\frac{\pi}{6} - d) + \sin(\frac{\pi}{6} + d)] = 3 + 2\sqrt{3}(\frac{1}{2}\cos d - \frac{\sqrt{3}}{2}\sin d + \frac{1}{2}\cos d + \frac{\sqrt{3}}{2}\sin d)$,

$= 3 + 2\sqrt{3}\cos d$,

当 $d = 0$, 即 $B = C = \frac{\pi}{6}$ 时, $\triangle ABC$ 的周长取得最大值 $3 + 2\sqrt{3}$.

另解: $a = 3$, $A = \frac{2\pi}{3}$, 又 $a^2 = b^2 + c^2 - 2bc\cos A$,

$\therefore 9 = b^2 + c^2 + bc = (b + c)^2 - bc \geq (b + c)^2 - \frac{1}{4}(b + c)^2$,

由 $b + c > 3$, 则 $b + c \leq 2\sqrt{3}$ (当且仅当 $b = c$ 时, “=” 成立),

则 $\triangle ABC$ 周长的最大值为 $3 + 2\sqrt{3}$.

□

28. (2020 • 浙江) 在锐角 $\triangle ABC$ 中, 角 A, B, C 所对的边分别为 a, b, c . 已知 $2b\sin A - \sqrt{3}a = 0$.

(I) 求角 B 的大小;

(II) 求 $\cos A + \cos B + \cos C$ 的取值范围.

解答 【解答】 解: (I) $\because 2b\sin A = \sqrt{3}a$,

$\therefore 2\sin B \sin A = \sqrt{3}\sin A$,

$\because \sin A \neq 0$,

$\therefore \sin B = \frac{\sqrt{3}}{2}$,

$\because \triangle ABC$ 为锐角三角形,

$\therefore B = \frac{\pi}{3}$,

(II) $\because \triangle ABC$ 为锐角三角形, $B = \frac{\pi}{3}$,

$\therefore C = \frac{2\pi}{3} - A$,

$\therefore \cos A + \cos B + \cos C = \cos A + \cos(\frac{2\pi}{3} - A) + \cos \frac{\pi}{3} = \cos A - \frac{1}{2}\cos A + \frac{\sqrt{3}}{2}\sin A + \frac{1}{2} = \frac{1}{2}$

$\cos A + \frac{\sqrt{3}}{2}\sin A + \frac{1}{2} = \sin(A + \frac{\pi}{6}) + \frac{1}{2}$,

$\triangle ABC$ 为锐角三角形, $0 < A < \frac{\pi}{2}$, $0 < C < \frac{\pi}{2}$,

解得 $\frac{\pi}{6} < A < \frac{\pi}{2}$,

$$\therefore \frac{\pi}{3} < A + \frac{\pi}{6} < \frac{2\pi}{3},$$

$$\therefore \frac{\sqrt{3}}{2} < \sin(A + \frac{\pi}{6}) \leq 1,$$

$$\therefore \frac{\sqrt{3}}{2} + \frac{1}{2} < \sin(A + \frac{\pi}{6}) + \frac{1}{2} \leq \frac{3}{2},$$

$$\therefore \cos A + \cos B + \cos C \text{ 的取值范围为 } (\frac{\sqrt{3}+1}{2}, \frac{3}{2}].$$

□

29. (2020 • 新课标 II) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\cos^2(\frac{\pi}{2} + A) + \cos A = \frac{5}{4}$.

(1) 求 A ;

(2) 若 $b - c = \frac{\sqrt{3}}{3}a$, 证明: $\triangle ABC$ 是直角三角形.

解答 【解答】解: (1) $\because \cos^2(\frac{\pi}{2} + A) + \cos A = \sin^2 A + \cos A = 1 - \cos^2 A + \cos A = \frac{5}{4}$,

$$\therefore \cos^2 A - \cos A + \frac{1}{4} = 0, \text{ 解得 } \cos A = \frac{1}{2},$$

$$\therefore A \in (0, \pi),$$

$$\therefore A = \frac{\pi}{3};$$

(2) 证明: $\because b - c = \frac{\sqrt{3}}{3}a, A = \frac{\pi}{3}$,

$$\therefore \text{由正弦定理可得 } \sin B - \sin C = \frac{\sqrt{3}}{3} \sin A = \frac{1}{2},$$

$$\therefore \sin B - \sin(\frac{2\pi}{3} - B) = \sin B - \frac{\sqrt{3}}{2} \cos B - \frac{1}{2} \sin B = \frac{1}{2} \sin B - \frac{\sqrt{3}}{2} \cos B = \sin(B - \frac{\pi}{3}) = \frac{1}{2},$$

$$\therefore B \in (0, \frac{2\pi}{3}), B - \frac{\pi}{3} \in (-\frac{\pi}{3}, \frac{\pi}{3}),$$

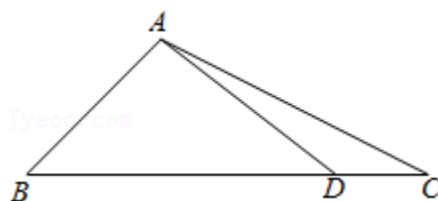
$$\therefore B - \frac{\pi}{3} = \frac{\pi}{6}, \text{ 可得 } B = \frac{\pi}{2}, \text{ 可得 } \triangle ABC \text{ 是直角三角形, 得证.}$$

□

30. (2020 • 江苏) 在 $\triangle ABC$ 中, 角 A, B, C 的对边分别为 a, b, c . 已知 $a = 3, c = \sqrt{2}, B = 45^\circ$.

(1) 求 $\sin C$ 的值;

(2) 在边 BC 上取一点 D , 使得 $\cos \angle ADC = -\frac{4}{5}$, 求 $\tan \angle DAC$ 的值.



解答 【解答】解: (1) 因为 $a = 3, c = \sqrt{2}, B = 45^\circ$, 由余弦定理可得: $b = \sqrt{a^2 + c^2 - 2ac \cos B} = \sqrt{9 + 2 - 2 \cdot 3\sqrt{2} \cdot \frac{\sqrt{2}}{2}} = \sqrt{5}$,

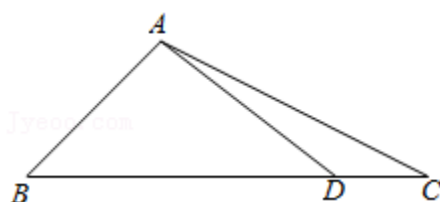
$$\text{由正弦定理可得 } \frac{c}{\sin C} = \frac{b}{\sin B}, \text{ 所以 } \sin C = \frac{c}{b} \cdot \sin 45^\circ = \frac{\sqrt{2}}{\sqrt{5}} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{5}}{5},$$

$$\text{所以 } \sin C = \frac{\sqrt{5}}{5};$$

(2) 因为 $\cos \angle ADC = -\frac{4}{5}$, 所以 $\sin \angle ADC = \sqrt{1 - \cos^2 \angle ADC} = \frac{3}{5}$,
 在三角形 ADC 中, 易知 C 为锐角, 由 (1) 可得 $\cos C = \sqrt{1 - \sin^2 C} = \frac{2\sqrt{5}}{5}$,
 所以在三角形 ADC 中,

$$\begin{aligned}\sin \angle DAC &= \sin(\angle ADC + C) \\ &= \sin \angle ADC \cos C + \cos \angle ADC \sin C = \frac{2\sqrt{5}}{25}.\end{aligned}$$

因为 $\angle DAC \in (0, \frac{\pi}{2})$, 所以 $\cos \angle DAC = \sqrt{1 - \sin^2 \angle DAC} = \frac{11\sqrt{5}}{25}$,
 所以 $\tan \angle DAC = \frac{\sin \angle DAC}{\cos \angle DAC} = \frac{2}{11}$.



□

31. (2020 • 山东) 在① $ac = \sqrt{3}$, ② $c \sin A = 3$, ③ $c = \sqrt{3}b$ 这三个条件中任选一个, 补充在下面问题中, 若问题中的三角形存在, 求 c 的值; 若问题中的三角形不存在, 说明理由.

问题: 是否存在 $\triangle ABC$, 它的内角 A, B, C 的对边分别为 a, b, c , 且 $\sin A = \sqrt{3} \sin B$, $C = \frac{\pi}{6}$, _____?

注: 如果选择多个条件分别解答, 按第一个解答计分.

解答 【解答】解: ① $ac = \sqrt{3}$.

$\triangle ABC$ 中, $\sin A = \sqrt{3} \sin B$, 即 $b = \frac{\sqrt{3}}{3}a$,

$$ac = \sqrt{3}, \therefore c = \frac{\sqrt{3}}{4},$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{a^2 + \frac{a^2}{3} - \frac{3}{4}}{A^2} = \frac{\sqrt{3}}{2},$$

$$\therefore a = \sqrt{3}, b = 1, c = 1.$$

② $c \sin A = 3$.

$\triangle ABC$ 中, $c \sin A = a \sin C = a \sin \frac{\pi}{6} = 3$, $\therefore a = 6$.

$\therefore \sin A = \sqrt{3} \sin B$, 即 $a = \sqrt{3}b$, $\therefore b = 2\sqrt{3}$.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{36 + 12 - c^2}{2 \times 6 \times 2\sqrt{3}} = \frac{\sqrt{3}}{2},$$

$$\therefore c = 2\sqrt{3}.$$

③ $c = \sqrt{3}b$.

$\therefore \sin A = \sqrt{3} \sin B$, 即 $a = \sqrt{3}b$,

又 $\therefore c = \sqrt{3}b$,

$$\cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{\sqrt{3}}{6} \neq \cos \frac{\pi}{6},$$

与已知条件 $C = \frac{\pi}{6}$ 相矛盾, 所以问题中的三角形不存在.

□

32. (2020 • 新课标 I) $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c . 已知 $B = 150^\circ$.

(1) 若 $a = \sqrt{3}c, b = 2\sqrt{7}$, 求 $\triangle ABC$ 的面积;

(2) 若 $\sin A + \sqrt{3}\sin C = \frac{\sqrt{2}}{2}$, 求 C .

解答 【解答】解: (1) $\triangle ABC$ 中, $B = 150^\circ, a = \sqrt{3}c, b = 2\sqrt{7}$,

$$\cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{3c^2+c^2-28}{2\sqrt{3}c^2} = -\frac{\sqrt{3}}{2},$$

$$\therefore c = 2 \text{ (负值舍去)}, a = 2\sqrt{3},$$

$$\therefore S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{1}{2} \cdot 2\sqrt{3} \cdot 2 \cdot \frac{1}{2} = \sqrt{3}.$$

$$(2) \sin A + \sqrt{3}\sin C = \frac{\sqrt{2}}{2},$$

$$\text{即 } \sin(180^\circ - 150^\circ - C) + \sqrt{3}\sin C = \frac{\sqrt{2}}{2},$$

$$\text{化简得 } \frac{1}{2}\cos C + \frac{\sqrt{3}}{2}\sin C = \frac{\sqrt{2}}{2},$$

$$\sin(C+30^\circ) = \frac{\sqrt{2}}{2},$$

$$\therefore 0^\circ < C < 30^\circ,$$

$$\therefore 30^\circ < C+30^\circ < 60^\circ,$$

$$\therefore C+30^\circ = 45^\circ,$$

$$\therefore C = 15^\circ.$$

□

33. (2020 • 北京) 在 $\triangle ABC$ 中, $a+b = 11$, 再从条件①、条件②这两个条件中选择一个作为已知, 求:

(I) a 的值;

(II) $\sin C$ 和 $\triangle ABC$ 的面积.

条件①: $c = 7, \cos A = -\frac{1}{7}$;

条件②: $\cos A = \frac{1}{8}, \cos B = \frac{9}{16}$.

注: 如果选择条件①和条件②分别解答, 按第一个解答计分.

解答 【解答】解: 选择条件① (I) 由余弦定理得 $a^2 = b^2 + c^2 - 2bccosA$, 即 $a^2 - b^2 = 49 - 14b \times (-\frac{1}{7}) = 49 + 2b$,

$$\therefore (a+b)(a-b) = 49 + 2b,$$

$$\therefore a+b = 11,$$

$$\therefore 11a - 11b = 49 + 2b,$$

$$\text{即 } 11a - 13b = 49,$$

$$\text{联立 } \begin{cases} a+b=11, \\ 11a-13b=49, \end{cases}, \text{ 解得 } a=8, b=3,$$

故 $a=8$.

(II) 在 $\triangle ABC$ 中, $\sin A > 0$,

$$\therefore \sin A = \sqrt{1 - \cos^2 A} = \frac{4\sqrt{3}}{7},$$

由正弦定理可得 $\frac{a}{\sin A} = \frac{c}{\sin C}$,

$$\therefore \sin C = \frac{c \sin A}{a} = \frac{7 \times \frac{4\sqrt{3}}{7}}{8} = \frac{\sqrt{3}}{2},$$

$$\therefore S_{\triangle ABC} = \frac{1}{2} ab \sin C = \frac{1}{2} \times 8 \times 3 \times \frac{\sqrt{3}}{2} = 6\sqrt{3}.$$

选择条件② (I) 在 $\triangle ABC$ 中, $\sin A > 0$, $\sin B > 0$, $C = \pi - (A+B)$,

$$\therefore \cos A = \frac{1}{8}, \cos B = \frac{9}{16},$$

$$\therefore \sin A = \sqrt{1 - \cos^2 A} = \frac{3\sqrt{7}}{8}, \sin B = \sqrt{1 - \cos^2 B} = \frac{5\sqrt{7}}{16},$$

由正弦定理可得 $\frac{a}{\sin A} = \frac{b}{\sin B}$,

$$\therefore \frac{a}{b} = \frac{\sin A}{\sin B} = \frac{6}{5},$$

$$\therefore a+b=11,$$

$$\therefore a=6, b=5,$$

故 $a=6$;

(II) 在 $\triangle ABC$ 中, $C = \pi - (A+B)$,

$$\therefore \sin C = \sin (A+B) = \sin A \cos B + \cos A \sin B = \frac{3\sqrt{7}}{8} \times \frac{9}{16} + \frac{5\sqrt{7}}{16} \times \frac{1}{8} = \frac{\sqrt{7}}{4},$$

$$\therefore S_{\triangle ABC} = \frac{1}{2} ab \sin C = \frac{1}{2} \times 6 \times 5 \times \frac{\sqrt{7}}{4} = \frac{15\sqrt{7}}{4}$$

□

34. (2020 • 天津) 在 $\triangle ABC$ 中, 角 A, B, C 所对的边分别为 a, b, c . 已知 $a = 2\sqrt{2}$, $b = 5$, $c = \sqrt{13}$.

(I) 求角 C 的大小;

(II) 求 $\sin A$ 的值;

(III) 求 $\sin (2A + \frac{\pi}{4})$ 的值.

解答 【解答】解: (I) 由余弦定理以及 $a = 2\sqrt{2}$, $b = 5$, $c = \sqrt{13}$,

$$\text{则 } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{8 + 25 - 13}{2 \times 2\sqrt{2} \times 5} = \frac{\sqrt{2}}{2},$$

$$\therefore C \in (0, \pi),$$

$$\therefore C = \frac{\pi}{4};$$

$$\begin{aligned} \text{(II) 由正弦定理, 以及 } C = \frac{\pi}{4}, a = 2\sqrt{2}, c = \sqrt{13}, \text{ 可得 } \sin A &= \frac{a \sin C}{c} = \frac{2\sqrt{2} \times \frac{\sqrt{2}}{2}}{\sqrt{13}} \\ &= \frac{2\sqrt{13}}{13}; \end{aligned}$$

$$\text{(III) 由 } a < c, \text{ 及 } \sin A = \frac{2\sqrt{13}}{13}, \text{ 可得 } \cos A = \sqrt{1 - \sin^2 A} = \frac{3\sqrt{13}}{13},$$

$$\text{则 } \sin 2A = 2 \sin A \cos A = 2 \times \frac{2\sqrt{13}}{13} \times \frac{3\sqrt{13}}{13} = \frac{12}{13},$$

$$\therefore \cos 2A = 2\cos^2 A - 1 = \frac{5}{13},$$

$$\therefore \sin\left(2A + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}(\sin 2A + \cos 2A) = \frac{\sqrt{2}}{2}\left(\frac{12}{13} + \frac{5}{13}\right) = \frac{17\sqrt{2}}{26}.$$

□

35. (2020 • 全国) 设 $\triangle ABC$ 的面积为 $10\sqrt{3}$, 内角 A, B, C 的对边分别为 a, b, c , 且 $a = 7$, $\sin^2 A = \sin^2 B + \sin^2 C - \sin B \sin C$. 求 A, b 和 c .

解答 【解答】解: 因为 $\sin^2 B + \sin^2 C - \sin^2 A = \sin B \sin C$,

所以 $b^2 + c^2 - a^2 = bc$,

$$\text{则 } \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{bc}{2bc} = \frac{1}{2},$$

因为 $0 < A < \pi$,

所以 $A = \frac{\pi}{3}$.

因为 $\triangle ABC$ 的面积为 $10\sqrt{3}$,

$$\text{所以 } \frac{1}{2}bc\sin A = \frac{\sqrt{3}}{4}bc = 10\sqrt{3}, \text{ 即 } bc = 40, \text{ ①}$$

又因为由余弦定理可得: $b^2 + c^2 - a^2 = bc$, $a = 7$,

$$\text{所以 } b^2 + c^2 = 89, \text{ ②}$$

所以由①②联立解得 $b = 5, c = 8$ 或 $b = 8, c = 5$.

□

36. (2021 • 上海) 已知 A, B, C 为 $\triangle ABC$ 的三个内角, a, b, c 是其三条边, $a = 2$, $\cos C = -\frac{1}{4}$.

(1) 若 $\sin A = 2\sin B$, 求 b, c ;

(2) 若 $\cos\left(A - \frac{\pi}{4}\right) = \frac{4}{5}$, 求 c .

解答 【解答】解: (1) 因为 $\sin A = 2\sin B$, 可得 $a = 2b$,

又 $a = 2$, 可得 $b = 1$,

$$\text{由于 } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{2^2 + 1^2 - c^2}{2 \times 2 \times 1} = -\frac{1}{4}, \text{ 可得 } c = \sqrt{6}.$$

$$(2) \text{ 因为 } \cos\left(A - \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}(\cos A + \sin A) = \frac{4}{5},$$

$$\text{可得 } \cos A + \sin A = \frac{4\sqrt{2}}{5},$$

$$\text{又 } \cos^2 A + \sin^2 A = 1,$$

$$\text{可解得 } \cos A = \frac{7\sqrt{2}}{10}, \sin A = \frac{\sqrt{2}}{10}, \text{ 或 } \sin A = \frac{7\sqrt{2}}{10}, \cos A = \frac{\sqrt{2}}{10},$$

因为 $\cos C = -\frac{1}{4}$, 可得 $\sin C = \frac{\sqrt{15}}{4}$, $\tan C = -\sqrt{15}$, 可得 C 为钝角,

若 $\sin A = \frac{7\sqrt{2}}{10}$, $\cos A = \frac{\sqrt{2}}{10}$, 可得 $\tan A = 7$, 可得 $\tan B = -\tan(A + C) =$

$$\frac{\tan A + \tan C}{\tan A \tan C - 1} = \frac{7 - \sqrt{15}}{7 \times (-\sqrt{15}) - 1} < 0,$$

可得 B 为钝角, 这与 C 为钝角矛盾, 舍去,

$$\text{所以 } \sin A = \frac{\sqrt{2}}{10}, \text{ 由正弦定理 } \frac{2}{\sin A} = \frac{c}{\sin C}, \text{ 可得 } c = \frac{5\sqrt{30}}{2}.$$

□

37. (2021 • 新高考 I) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c . 已知 $b^2 = ac$, 点 D 在边 AC 上, $BD \sin \angle ABC = a \sin C$.

(1) 证明: $BD = b$;

(2) 若 $AD = 2DC$, 求 $\cos \angle ABC$.

解答 【解答】解: (1) 证明: 由正弦定理知, $\frac{b}{\sin \angle ABC} = \frac{c}{\sin \angle ACB} = 2R$,

$$\therefore b = 2R \sin \angle ABC, c = 2R \sin \angle ACB,$$

$$\therefore b^2 = ac, \therefore b \cdot 2R \sin \angle ABC = a \cdot 2R \sin \angle ACB,$$

$$\text{即 } b \sin \angle ABC = a \sin C,$$

$$\therefore BD \sin \angle ABC = a \sin C,$$

$$\therefore BD = b;$$

(2) 法一: 由 (1) 知 $BD = b$,

$$\therefore AD = 2DC, \therefore AD = \frac{2}{3}b, DC = \frac{1}{3}b,$$

$$\text{在 } \triangle ABD \text{ 中, 由余弦定理知, } \cos \angle BDA = \frac{BD^2 + AD^2 - AB^2}{2BD \cdot AD} = \frac{b^2 + (\frac{2}{3}b)^2 - c^2}{2b \cdot \frac{2}{3}b} = \frac{13b^2 - 9c^2}{12b^2},$$

,

$$\text{在 } \triangle CBD \text{ 中, 由余弦定理知, } \cos \angle BDC = \frac{BD^2 + CD^2 - BC^2}{2BD \cdot CD} = \frac{b^2 + (\frac{1}{3}b)^2 - a^2}{2b \cdot \frac{1}{3}b} = \frac{10b^2 - 9a^2}{6b^2},$$

,

$$\therefore \angle BDA + \angle BDC = \pi,$$

$$\therefore \cos \angle BDA + \cos \angle BDC = 0,$$

$$\text{即 } \frac{13b^2 - 9c^2}{12b^2} + \frac{10b^2 - 9a^2}{6b^2} = 0,$$

$$\text{得 } 11b^2 = 3c^2 + 6a^2,$$

$$\therefore b^2 = ac,$$

$$\therefore 3c^2 - 11ac + 6a^2 = 0,$$

$$\therefore c = 3a \text{ 或 } c = \frac{2}{3}a,$$

$$\text{在 } \triangle ABC \text{ 中, 由余弦定理知, } \cos \angle ABC = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + c^2 - ac}{2ac},$$

$$\text{当 } c = 3a \text{ 时, } \cos \angle ABC = \frac{7}{6} > 1 \text{ (舍);}$$

$$\text{当 } c = \frac{2}{3}a \text{ 时, } \cos \angle ABC = \frac{7}{12};$$

$$\text{综上所述, } \cos \angle ABC = \frac{7}{12}.$$

法二: $\because$ 点 D 在边 AC 上且 $AD = 2DC$,

$$\therefore \overrightarrow{BD} = \frac{1}{3}\overrightarrow{BA} + \frac{2}{3}\overrightarrow{BC},$$

$$\therefore \overrightarrow{BD}^2 = \frac{1}{3}\overrightarrow{BA} \cdot \overrightarrow{BD} + \frac{2}{3}\overrightarrow{BC} \cdot \overrightarrow{BD},$$

而由 (1) 知 $BD = b$,

$$\therefore b^2 = \frac{1}{3}bc \cos \angle ABD + \frac{2}{3}ab \cos \angle CBD,$$

$$\text{即 } 3b = c \cdot \cos \angle ABD + 2a \cdot \cos \angle CBD,$$

$$\text{由余弦定理知: } 3b = c \cdot \frac{b^2 + c^2 - \frac{4}{9}b^2}{2bc} + 2a \cdot \frac{a^2 + b^2 - \frac{1}{9}b^2}{2ab},$$

$$\therefore 11b^2 = 3c^2 + 6a^2,$$

$$\therefore b^2 = ac,$$

$$\therefore 3c^2 - 11ac + 6a^2 = 0,$$

$$\therefore c = 3a \text{ 或 } c = \frac{2}{3}a,$$

$$\text{在 } \triangle ABC \text{ 中, 由余弦定理知, } \cos \angle ABC = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + c^2 - ac}{2ac},$$

$$\text{当 } c = 3a \text{ 时, } \cos \angle ABC = \frac{7}{6} > 1 \text{ (舍);}$$

$$\text{当 } c = \frac{2}{3}a \text{ 时, } \cos \angle ABC = \frac{7}{12};$$

$$\text{综上所述, } \cos \angle ABC = \frac{7}{12}.$$

$$\text{法三: 在 } \triangle BCD \text{ 中, 由正弦定理可知 } a \sin C = BD \sin \angle BDC = b \sin \angle BDC,$$

$$\text{而由题意可知 } ac = b^2 \Rightarrow a \sin C = b \sin \angle ABC,$$

$$\text{于是 } \sin \angle BDC = \sin \angle ABC, \text{ 从而 } \angle BDC = \angle ABC \text{ 或 } \angle BDC + \angle ABC = \pi.$$

$$\text{若 } \angle BDC = \angle ABC, \text{ 则 } \triangle CBD \sim \triangle CAB, \text{ 于是 } CB^2 = CD \cdot CA \Rightarrow a^2 = \frac{b^2}{3} \Rightarrow a:$$

$$b: c = 1: \sqrt{3}: 3,$$

无法构成三角形, 不合题意.

$$\text{若 } \angle BDC + \angle ABC = \pi, \text{ 则 } \angle ADB = \angle ABC \Rightarrow \triangle ABD \sim \triangle ACB,$$

$$\text{于是 } AB^2 = AD \cdot AC \Rightarrow c^2 = \frac{2b^2}{3} \Rightarrow a: b: c = 3: \sqrt{6}: 2, \text{ 满足题意,}$$

$$\text{因此由余弦定理可得 } \cos \angle ABC = \frac{a^2 + c^2 - b^2}{2ac} = \frac{7}{12}.$$

□

38. (2021 • 上海) 在 $\triangle ABC$ 中, 已知 $a = 3$, $b = 2c$.

(1) 若 $A = \frac{2\pi}{3}$, 求 $S_{\triangle ABC}$.

(2) 若 $2\sin B - \sin C = 1$, 求 $\triangle ABC$ 的周长.

$$\text{解答 【解答】解: (1) 由余弦定理得 } \cos A = -\frac{1}{2} = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5c^2 - 9}{4c^2},$$

$$\text{解得 } c^2 = \frac{9}{7},$$

$$\therefore S_{\triangle ABC} = \frac{1}{2}bc \sin A = \frac{\sqrt{3}}{4} \cdot 2c^2 = \frac{9\sqrt{3}}{14};$$

$$(2) \because b = 2c, \therefore \text{由正弦定理得 } \sin B = 2\sin C, \text{ 又 } \because 2\sin B - \sin C = 1,$$

$$\therefore \sin C = \frac{1}{3}, \sin B = \frac{2}{3}, \therefore \sin C < \sin B, \therefore C < B, \therefore C \text{ 为锐角,}$$

$$\therefore \cos C = \sqrt{1 - \left(\frac{1}{3}\right)^2} = \frac{2\sqrt{2}}{3}.$$

$$\text{由余弦定理得: } c^2 = a^2 + b^2 - 2ab \cos C, \text{ 又 } \because a = 3, b = 2c,$$

$$\therefore c^2 = 9 + 4c^2 - 8\sqrt{2}c, \text{ 得: } 3c^2 - 8\sqrt{2}c + 9 = 0, \text{ 解得: } c = \frac{4\sqrt{2} \pm \sqrt{5}}{3}.$$

$$\text{当 } c = \frac{4\sqrt{2} + \sqrt{5}}{3} \text{ 时, } b = \frac{8\sqrt{2} + 2\sqrt{5}}{3}, \text{ 此时 } \triangle ABC \text{ 的周长为 } 3 + 4\sqrt{2} + \sqrt{5};$$

$$\text{当 } c = \frac{4\sqrt{2} - \sqrt{5}}{3} \text{ 时, } b = \frac{8\sqrt{2} - 2\sqrt{5}}{3}, \text{ 此时 } \triangle ABC \text{ 的周长为 } 3 + 4\sqrt{2} - \sqrt{5}.$$

□

39. (2021 • 天津) 在 $\triangle ABC$ 中, 内角 A, B, C 的对边分别为 a, b, c , 且 $\sin A: \sin B: \sin C = 2: 1: \sqrt{2}$, $b = \sqrt{2}$.

(1) 求 a 的值;

(2) 求 $\cos C$ 的值;

(3) 求 $\sin(2C - \frac{\pi}{6})$ 的值.

解答 【解答】解: (1) $\because \triangle ABC$ 中, $\sin A : \sin B : \sin C = 2 : 1 : \sqrt{2}$, $\therefore a : b : c = 2 : 1 : \sqrt{2}$,

$\therefore b = \sqrt{2}$, $\therefore a = 2b = 2\sqrt{2}$, $c = \sqrt{2}b = 2$.

(2) $\triangle ABC$ 中, 由余弦定理可得 $\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{8 + 2 - 4}{2 \cdot 2\sqrt{2} \cdot \sqrt{2}} = \frac{3}{4}$.

(3) 由 (2) 可得 $\sin C = \sqrt{1 - \cos^2 C} = \frac{\sqrt{7}}{4}$,

$\therefore \sin 2C = 2\sin C \cos C = \frac{3\sqrt{7}}{8}$, $\cos 2C = 2\cos^2 C - 1 = \frac{1}{8}$,

$\sin(2C - \frac{\pi}{6}) = \sin 2C \cos \frac{\pi}{6} - \cos 2C \sin \frac{\pi}{6} = \frac{3\sqrt{21} - 1}{16}$.

□

40. (2021 • 北京) 在 $\triangle ABC$ 中, $c = 2b \cos B$, $\angle C = \frac{2\pi}{3}$.

(I) 求 $\angle B$;

(II) 再在条件①、条件②、条件③这三个条件中选择一个作为已知, 使 $\triangle ABC$ 存在且唯一确定, 并求 BC 边上的中线的长.

条件① $c = \sqrt{2}b$;

条件② $\triangle ABC$ 的周长为 $4 + 2\sqrt{3}$;

条件③ $\triangle ABC$ 的面积为 $\frac{3\sqrt{3}}{4}$.

注: 如果选择的条件不符合要求, 第 (II) 问得 0 分; 如果选择多个符合要求的条件分别解答, 按第一个解答计分.

解答 【解答】解: (I) $\because c = 2b \cos B$,

由正弦定理可得 $\sin C = 2\sin B \cos B$, 即 $\sin C = \sin 2B$,

$\therefore C = \frac{2\pi}{3}$,

$\therefore$ 当 $C = 2B$ 时, $B = \frac{\pi}{3}$, 即 $C + B = \pi$, 不符合题意, 舍去,

$\therefore C + 2B = \pi$,

$\therefore 2B = \frac{\pi}{3}$,

即 $B = \frac{\pi}{6}$.

(II) 选① $c = \sqrt{2}b$,

由正弦定理可得

$\frac{c}{b} = \frac{\sin C}{\sin B} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$, 与已知条件 $c = \sqrt{2}b$ 矛盾, 故 $\triangle ABC$ 不存在,

选② 周长为 $4 + 2\sqrt{3}$,

$\therefore C = \frac{2\pi}{3}$, $B = \frac{\pi}{6}$,

$\therefore A = \frac{\pi}{6}$,

由正弦定理可得 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, 即 $\frac{a}{\frac{1}{2}} = \frac{b}{\frac{1}{2}} = \frac{c}{\frac{\sqrt{3}}{2}} = 2R$,

$\therefore a = R$, $b = R$, $c = \sqrt{3}R$,

$$\therefore a+b+c = (2+\sqrt{3})R = 4+2\sqrt{3},$$

$$\therefore R = 2, \text{ 即 } a = 2, b = 2, c = 2\sqrt{3},$$

$\therefore \triangle ABC$ 存在且唯一确定,

设 BC 的中点为 D,

$$\therefore CD = 1,$$

在 $\triangle ACD$ 中, 运用余弦定理, $AD^2 = AC^2 + CD^2 - 2AC \cdot CD \cdot \cos \angle C$,

$$\text{即 } AD^2 = 4 + 1 - 2 \cdot 2 \cdot 1 \left(-\frac{1}{2}\right) = 7, AD = \sqrt{7},$$

$\therefore BC$ 边上的中线的长度 $\sqrt{7}$.

$$\text{选③面积为 } S_{\triangle ABC} = \frac{3\sqrt{3}}{4},$$

$$\therefore A = B = \frac{\pi}{6},$$

$$\therefore a = b,$$

$$\therefore S_{\triangle ABC} = \frac{1}{2}ab \sin C = \frac{1}{2}a^2 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}, \text{ 解得 } a = \sqrt{3},$$

余弦定理可得

$$AD^2 = AC^2 + CD^2 - 2 \times AC \times CD \times \cos \frac{2\pi}{3} = 3 + \frac{3}{4} + \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{21}{4},$$

$$AD = \frac{\sqrt{21}}{2}.$$

□

41. (2021 • 全国) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c. 已知 $a = 2\sqrt{6}$, $b = 3$, $\sin^2(B+C) + \sqrt{2}\sin 2A = 0$, 求 c 及 $\cos B$.

解答 【解答】解: $\because A+B+C = \pi$, $\sin^2(B+C) + \sqrt{2}\sin 2A = 0$,

$$\therefore \sin^2 A + 2\sqrt{2}\sin A \cos A = 0,$$

$$\therefore \sin A \neq 0,$$

$$\therefore \sin A = -2\sqrt{2}\cos A,$$

$$\therefore \cos A < 0, A \text{ 为钝角},$$

$$\text{又 } \therefore \sin^2 A + \cos^2 A = 1,$$

$$\therefore \cos A = \frac{1}{3}, \sin A = \frac{2\sqrt{2}}{3},$$

由正弦定理可得, $\frac{a}{\sin A} = \frac{b}{\sin B}$, 即 $\frac{2\sqrt{6}}{\frac{2\sqrt{2}}{3}} = \frac{3}{\sin B}$, 解得 $\sin B = \frac{\sqrt{3}}{3}$,

又 $\therefore \sin^2 B + \cos^2 B = 1$, B 为锐角,

$$\therefore \cos B = \frac{\sqrt{6}}{3},$$

$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \text{ 即 } \frac{24 + c^2 - 9}{4\sqrt{6}c} = \frac{\sqrt{6}}{3}, \text{ 化简整理可得, } c^2 - 8c + 15 = 0, \text{ 解得 } c = 3$$

或 $c = 5$,

$$\therefore \angle A > \angle C,$$

$$\therefore a > c,$$

$$\therefore 5 > 2\sqrt{6},$$

$$\therefore c = 3,$$

$$\text{故 } c = 3, \cos B = \frac{\sqrt{6}}{3}.$$

□

42. (2022 • 乙卷) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin C \sin(A-B) = \sin B \sin(C-A)$.

(1) 证明: $2a^2 = b^2 + c^2$;

(2) 若 $A = 2B$, 求 C ;

(3) 若 $a = 5$, $\cos A = \frac{25}{31}$, 求 $\triangle ABC$ 的周长.

解答 【解答】(1) 解法一 (恒等变换与正弦定理): 展开原式并移项, 得

$$\sin A(\sin B \cos C + \cos B \sin C) = 2 \cos A \sin B \sin C.$$

由 $\sin(B+C) = \sin A$, 可得

$$\sin^2 A = 2 \cos A \sin B \sin C.$$

由正弦定理, $a^2 = 2bc \cos A$; 再结合余弦定理 $a^2 = b^2 + c^2 - 2bc \cos A$, 得到

$$2a^2 = b^2 + c^2.$$

解法二 (边角互化与余弦定理): 展开原式后利用正弦定理将正弦值换成边, 得

$$ac \cos B - bc \cos A = bc \cos A - ab \cos C,$$

即 $ac \cos B = 2bc \cos A - ab \cos C$. 分别利用余弦定理表示三个余弦值, 化简同样得到

$$2a^2 = b^2 + c^2.$$

(2) 由 $A = 2B$, 原条件化为

$$\sin C \sin B = \sin B \sin(C-A).$$

因为 $\sin B \neq 0$, 所以 $\sin C = \sin(C-A)$. 结合三角形内角范围, 得 $2C-A = \pi$. 再由 $A = 2B$ 和 $A+B+C = \pi$, 解得

$$C = \frac{5\pi}{8}.$$

(3) 由 (1) 知 $b^2 + c^2 = 2a^2 = 50$, 且

$$2bc = \frac{a^2}{\cos A} = 31.$$

所以 $(b+c)^2 = b^2 + c^2 + 2bc = 81$, 从而 $b+c = 9$. 故 $\triangle ABC$ 的周长为

$$a+b+c = 14.$$

□

43. (2022 • 新高考 II) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 分别以 a, b, c 为边长的三个正三角形的面积依次为 S_1, S_2, S_3 . 已知 $S_1 - S_2 + S_3 = \frac{\sqrt{3}}{2}$, $\sin B = \frac{1}{3}$.

(1) 求 $\triangle ABC$ 的面积;

(2) 若 $\sin A \sin C = \frac{\sqrt{2}}{3}$, 求 b .

解答 【解答】解: (1) $S_1 = \frac{1}{2}a^2 \sin 60^\circ = \frac{\sqrt{3}}{4}a^2$,

$$S_2 = \frac{1}{2}b^2 \sin 60^\circ = \frac{\sqrt{3}}{4}b^2,$$

$$S_3 = \frac{1}{2}c^2 \sin 60^\circ = \frac{\sqrt{3}}{4}c^2,$$

$$\therefore S_1 - S_2 + S_3 = \frac{\sqrt{3}}{4}a^2 - \frac{\sqrt{3}}{4}b^2 + \frac{\sqrt{3}}{4}c^2 = \frac{\sqrt{3}}{2},$$

$$\text{解得: } a^2 - b^2 + c^2 = 2,$$

$$\therefore \sin B = \frac{1}{3}, a^2 - b^2 + c^2 = 2 > 0, \text{ 即 } \cos B > 0,$$

$$\therefore \cos B = \frac{2\sqrt{2}}{3},$$

$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{2\sqrt{2}}{3},$$

$$\text{解得: } ac = \frac{3\sqrt{2}}{4},$$

$$S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{\sqrt{2}}{8}.$$

$$\therefore \triangle ABC \text{ 的面积为 } \frac{\sqrt{2}}{8}.$$

$$(2) \text{ 由正弦定理得: } \frac{b}{\sin B} = \frac{a}{\sin A} = \frac{c}{\sin C},$$

$$\therefore a = \frac{b \sin A}{\sin B}, c = \frac{b \sin C}{\sin B},$$

$$\text{由 (1) 得 } ac = \frac{3\sqrt{2}}{4},$$

$$\therefore ac = \frac{b \sin A}{\sin B} \cdot \frac{b \sin C}{\sin B} = \frac{3\sqrt{2}}{4}$$

$$\text{已知, } \sin B = \frac{1}{3}, \sin A \sin C = \frac{\sqrt{2}}{3},$$

$$\text{解得: } b = \frac{1}{2}.$$

□

44. (2022 • 新高考 I) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\frac{\cos A}{1 + \sin A} = \frac{\sin 2B}{1 + \cos 2B}$.

(1) 若 $C = \frac{2\pi}{3}$, 求 B ;

(2) 求 $\frac{a^2 + b^2}{c^2}$ 的最小值.

解答 【解答】解: (1) $\therefore \frac{\cos A}{1 + \sin A} = \frac{\sin 2B}{1 + \cos 2B}$, $1 + \cos 2B = 2\cos^2 B \neq 0$, $\cos B \neq 0$.

$$\therefore \frac{\cos A}{1 + \sin A} = \frac{2 \sin B \cos B}{2 \cos^2 B} = \frac{\sin B}{\cos B},$$

$$\text{化为: } \cos A \cos B = \sin A \sin B + \sin B,$$

$$\therefore \cos (B + A) = \sin B,$$

$$\therefore -\cos C = \sin B, C = \frac{2\pi}{3},$$

$$\therefore \sin B = \frac{1}{2},$$

$$\therefore 0 < B < \frac{\pi}{3}, \therefore B = \frac{\pi}{6}.$$

(2) 由 (1) 可得: $-\cos C = \sin B > 0$, $\therefore \cos C < 0$, $C \in (\frac{\pi}{2}, \pi)$,

$\therefore C$ 为钝角, B, A 都为锐角, $B = C - \frac{\pi}{2}$.

$$\sin A = \sin(B+C) = \sin(2C - \frac{\pi}{2}) = -\cos 2C,$$

$$\frac{a^2+b^2}{c^2} = \frac{\sin^2 A + \sin^2 B}{\sin^2 C} = \frac{\cos^2 2C + \cos^2 C}{\sin^2 C} = \frac{(1-2\sin^2 C)^2 + (1-\sin^2 C)}{\sin^2 C} = \frac{2+4\sin^4 C-5\sin^2 C}{\sin^2 C} = \frac{2}{\sin^2 C} + 4\sin^2 C - 5 \geq 2\sqrt{2 \times 4} - 5 = 4\sqrt{2} - 5, \text{ 当且仅当 } \sin C = \frac{1}{\sqrt{2}} \text{ 时取等号.}$$

$$\therefore \frac{a^2+b^2}{c^2} \text{ 的最小值为 } 4\sqrt{2} - 5.$$

□

45. (2022 • 北京) 在 $\triangle ABC$ 中, $\sin 2C = \sqrt{3}\sin C$.

(I) 求 $\angle C$;

(II) 若 $b = 6$, 且 $\triangle ABC$ 的面积为 $6\sqrt{3}$, 求 $\triangle ABC$ 的周长.

解答 【解答】解: (I) $\therefore \sin 2C = \sqrt{3}\sin C$,

$$\therefore 2\sin C \cos C = \sqrt{3}\sin C,$$

$$\text{又 } \sin C \neq 0, \therefore 2\cos C = \sqrt{3},$$

$$\therefore \cos C = \frac{\sqrt{3}}{2}, \therefore 0 < C < \pi,$$

$$\therefore C = \frac{\pi}{6};$$

(II) $\therefore \triangle ABC$ 的面积为 $6\sqrt{3}$,

$$\therefore \frac{1}{2}ab\sin C = 6\sqrt{3},$$

$$\text{又 } b = 6, C = \frac{\pi}{6},$$

$$\therefore \frac{1}{2} \times a \times 6 \times \frac{1}{2} = 6\sqrt{3},$$

$$\therefore a = 4\sqrt{3},$$

$$\text{又 } \cos C = \frac{a^2+b^2-c^2}{2ab},$$

$$\therefore \frac{\sqrt{3}}{2} = \frac{(4\sqrt{3})^2 + 6^2 - c^2}{2 \times 4\sqrt{3} \times 6},$$

$$\therefore c = 2\sqrt{3},$$

$$\therefore a+b+c = 6+6\sqrt{3},$$

$$\therefore \triangle ABC \text{ 的周长为 } 6+6\sqrt{3}.$$

□

46. (2022 • 浙江) 在 $\triangle ABC$ 中, 角 A, B, C 所对的边分别为 a, b, c . 已知 $4a = \sqrt{5}c$, $\cos C = \frac{3}{5}$.

(I) 求 $\sin A$ 的值;

(II) 若 $b = 11$, 求 $\triangle ABC$ 的面积.

解答 【解答】解: (I) 因为 $\cos C = \frac{3}{5} > 0$, 所以 $C \in (0, \frac{\pi}{2})$, 且 $\sin C = \sqrt{1 - \cos^2 C} = \frac{4}{5}$,

由正弦定理可得: $\frac{a}{\sin A} = \frac{c}{\sin C}$,

即有 $\sin A = \frac{a \sin C}{c} = \frac{a}{c} \sin C = \frac{\sqrt{5}}{4} \times \frac{4}{5} = \frac{\sqrt{5}}{5}$;

(II) 因为 $4a = \sqrt{5}c \Rightarrow a = \frac{\sqrt{5}}{4}c < c$,

所以 $A < C$, 故 $A \in (0, \frac{\pi}{2})$,

又因为 $\sin A = \frac{\sqrt{5}}{5}$, 所以 $\cos A = \frac{2\sqrt{5}}{5}$,

所以 $\sin B = \sin[\pi - (A+C)] = \sin(A+C) = \sin A \cos C + \cos A \sin C = \frac{11\sqrt{5}}{25}$;

由正弦定理可得: $\frac{a}{\sin A} = \frac{c}{\sin C} = \frac{b}{\sin B} = 5\sqrt{5}$,

所以 $a = 5\sqrt{5} \sin A = 5$,

所以 $S_{\triangle ABC} = \frac{1}{2}ab \sin C = \frac{1}{2} \times 5 \times 11 \times \frac{4}{5} = 22$.

□

47. (2022 • 天津) 在 $\triangle ABC$ 中, 角 A, B, C 所对的边分别为 a, b, c . 已知 $a = \sqrt{6}$, $b = 2c$, $\cos A = -\frac{1}{4}$.

(1) 求 c 的值;

(2) 求 $\sin B$ 的值;

(3) 求 $\sin(2A-B)$ 的值.

解答 【解答】 解 (1) 因为 $a = \sqrt{6}$, $b = 2c$, $\cos A = -\frac{1}{4}$,

由余弦定理可得 $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{4c^2 + c^2 - 6}{4c^2} = -\frac{1}{4}$,

解得: $c = 1$;

(2) $\cos A = -\frac{1}{4}$, $A \in (0, \pi)$, 所以 $\sin A = \sqrt{1 - \cos^2 A} = \frac{\sqrt{15}}{4}$,

由 $b = 2c$, 可得 $\sin B = 2\sin C$,

由正弦定理可得 $\frac{a}{\sin A} = \frac{c}{\sin C}$, 即 $\frac{\sqrt{6}}{\frac{\sqrt{15}}{4}} = \frac{1}{\sin C}$,

可得 $\sin C = \frac{\sqrt{10}}{8}$,

所以 $\sin B = 2\sin C = 2 \times \frac{\sqrt{10}}{8} = \frac{\sqrt{10}}{4}$;

(3) 因为 $\cos A = -\frac{1}{4}$, $\sin A = \frac{\sqrt{15}}{4}$,

所以 $\sin 2A = 2\sin A \cos A = 2 \times (-\frac{1}{4}) \times \frac{\sqrt{15}}{4} = -\frac{\sqrt{15}}{8}$, $\cos 2A = 2\cos^2 A - 1 = 2 \times \frac{1}{16} - 1 = -\frac{7}{8}$,

$\sin B = \frac{\sqrt{10}}{4}$, 可得 $\cos B = \frac{\sqrt{6}}{4}$,

所以 $\sin(2A-B) = \sin 2A \cos B - \cos 2A \sin B = -\frac{\sqrt{15}}{8} \times \frac{\sqrt{6}}{4} - (-\frac{7}{8}) \times \frac{\sqrt{10}}{4} = \frac{\sqrt{10}}{8}$,

所以 $\sin(2A-B)$ 的值为 $\frac{\sqrt{10}}{8}$.

□

48. (2022 • 全国) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin A = 3\sin B$, $C = \frac{\pi}{3}$, $c = \sqrt{7}$.

(1) 求 a ;

(2) 求 $\sin A$.

解答 【解答】解：(1) $\because \sin A = 3\sin B$,

$\therefore$ 由正弦定理可得, $a = 3b$,

$\therefore$ 由余弦定理可得, $c^2 = a^2 + b^2 - 2ab\cos C$, 即 $7 = 9b^2 + b^2 - 3b^2$, 解得 $b = 1$,

$\therefore a = 3$.

(2) $\because a = 3, C = \frac{\pi}{3}, c = \sqrt{7}$,

$$\therefore \sin A = \frac{a \sin C}{c} = \frac{3 \cdot \frac{\sqrt{3}}{2}}{\sqrt{7}} = \frac{3\sqrt{21}}{14}.$$

□

49. (10 分) (2023 • 新高考 I) 已知在 $\triangle ABC$ 中, $A+B = 3C$, $2\sin(A-C) = \sin B$.

(1) 求 $\sin A$;

(2) 设 $AB = 5$, 求 AB 边上的高.

解答 【解答】解：(1) $\because A+B = 3C, A+B+C = \pi$,

$$\therefore 4C = \pi,$$

$$\therefore C = \frac{\pi}{4},$$

$$\because 2\sin(A-C) = \sin B,$$

$$\therefore 2\sin(A-C) = \sin[\pi - (A+C)] = \sin(A+C),$$

$$\therefore 2\sin A \cos C - 2\cos A \sin C = \sin A \cos C + \cos A \sin C,$$

$$\therefore \sin A \cos C = 3\cos A \sin C,$$

$$\therefore \frac{\sqrt{2}}{2} \sin A = 3 \cdot \frac{\sqrt{2}}{2} \cos A,$$

$$\therefore \sin A = 3\cos A, \text{ 即 } \cos A = \frac{1}{3}\sin A,$$

$$\text{又 } \because \sin^2 A + \cos^2 A = 1, \therefore \sin^2 A + \frac{1}{9}\sin^2 A = 1,$$

$$\text{解得 } \sin^2 A = \frac{9}{10},$$

$$\text{又 } \because A \in (0, \pi), \therefore \sin A > 0,$$

$$\therefore \sin A = \frac{3\sqrt{10}}{10};$$

$$(2) \text{ 由 (1) 可知 } \sin A = \frac{3\sqrt{10}}{10}, \cos A = \frac{1}{3}\sin A = \frac{\sqrt{10}}{10},$$

$$\therefore \sin B = \sin(A+C) = \sin A \cos C + \cos A \sin C = \frac{3\sqrt{10}}{10} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{10}}{10} \cdot \frac{\sqrt{2}}{2} = \frac{2\sqrt{5}}{5},$$

$$\therefore \frac{AB}{\sin C} = \frac{AC}{\sin B} = \frac{BC}{\sin A} = \frac{5}{\sin \frac{\pi}{4}} = 5\sqrt{2},$$

$$\therefore AC = 5\sqrt{2}\sin B = 5\sqrt{2} \times \frac{2\sqrt{5}}{5} = 2\sqrt{10}, BC = 5\sqrt{2}\sin A = 5\sqrt{2} \times \frac{3\sqrt{10}}{10} = 3\sqrt{5},$$

设 AB 边上的高为 h ,

$$\text{则 } \frac{1}{2}AB \cdot h = \frac{1}{2}AC \cdot BC \sin C,$$

$$\therefore \frac{5}{2}h = \frac{1}{2} \times 2\sqrt{10} \times 3\sqrt{5} \times \frac{\sqrt{2}}{2},$$

解得 $h = 6$,

即 AB 边上的高为 6.

□

50. (12 分) (2023 • 甲卷) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知

$$\frac{b^2+c^2-a^2}{\cos A} = 2.$$

(1) 求 bc ;

(2) 若 $\frac{a \cos B - b \cos A}{a \cos B + b \cos A} \cdot \frac{b}{c} = 1$, 求 $\triangle ABC$ 面积.

解答 【解答】解: (1) 因为 $\frac{b^2+c^2-a^2}{\cos A} = \frac{2bc \cos A}{\cos A} = 2bc = 2$,

所以 $bc = 1$;

$$(2) \frac{a \cos B - b \cos A}{a \cos B + b \cos A} \cdot \frac{b}{c} = \frac{\sin A \cos B - \sin B \cos A}{\sin A \cos B + \sin B \cos A} \cdot \frac{\sin B}{\sin C} = 1,$$

$$\text{所以 } \frac{\sin(A-B)}{\sin(A+B)} \cdot \frac{\sin B}{\sin C} = \frac{\sin(A-B) - \sin B}{\sin C} = 1,$$

$$\text{所以 } \sin(A-B) - \sin B = \sin C = \sin(A+B),$$

$$\text{所以 } \sin A \cos B - \sin B \cos A - \sin B = \sin A \cos B + \sin B \cos A,$$

$$\text{即 } \cos A = -\frac{1}{2},$$

$$\text{由 } A \text{ 为三角形内角得 } A = \frac{2\pi}{3},$$

$$\triangle ABC \text{ 面积 } S = \frac{1}{2}bc \sin A = \frac{1}{2} \cdot 1 \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}.$$

□

51. (10 分) (2023 • 新高考 II) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c ,

已知 $\triangle ABC$ 面积为 $\sqrt{3}$, D 为 BC 的中点, 且 $AD = 1$.

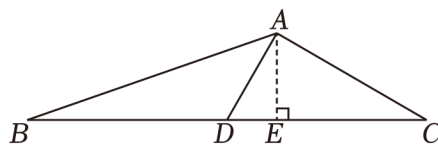
(1) 若 $\angle ADC = \frac{\pi}{3}$, 求 $\tan B$;

(2) 若 $b^2+c^2 = 8$, 求 b, c .

解答 【解答】解: (1) D 为 BC 中点, $S_{\triangle ABC} = \sqrt{3}$,

$$\text{则 } S_{\triangle ACD} = \frac{\sqrt{3}}{2},$$

过 A 作 $AE \perp BC$, 垂足为 E , 如图所示:



$$\triangle ADE \text{ 中, } DE = \frac{1}{2}, AE = \frac{\sqrt{3}}{2}, S_{\triangle ACD} = \frac{1}{2} \cdot \frac{\sqrt{3}}{2} CD = \frac{\sqrt{3}}{2}, \text{ 解得 } CD = 2,$$

$$\therefore BD = 2, BE = \frac{5}{2},$$

$$\text{故 } \tan B = \frac{AE}{BE} = \frac{\frac{\sqrt{3}}{2}}{\frac{5}{2}} = \frac{\sqrt{3}}{5};$$

$$(2) \overrightarrow{AD} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC}),$$

$$AD^2 = \frac{1}{4}(c^2 + b^2 + 2bc \cos A),$$

$$AD = 1, b^2+c^2 = 8,$$

$$\text{则 } 1 = \frac{1}{4}(8 + 2bc \cos A),$$

$$\therefore bc \cos A = -2 \quad \text{①},$$

$$S_{\triangle ABC} = \frac{1}{2}bc \sin A = \sqrt{3}, \text{ 即 } bc \sin A = 2\sqrt{3} \quad \text{②},$$

由①②解得 $\tan A = -\sqrt{3}$,

$$\therefore A = \frac{2\pi}{3},$$

$$\therefore bc = 4, \text{ 又 } b^2 + c^2 = 8,$$

$$\therefore b = c = 2.$$

□

52. (2024 全国 1) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin C = \sqrt{2} \cos B$, $a^2 + b^2 - c^2 = \sqrt{2}ab$

(1) 求 B ;

(2) 若 $\triangle ABC$ 的面积为 $3 + \sqrt{3}$, 求 c .

解答 【答案】 (1) $B = \frac{\pi}{3}$

(2) $2\sqrt{2}$

【解析】

【分析】 (1) 由余弦定理、平方关系依次求出 $\cos C, \sin C$, 最后结合已知 $\sin C = \sqrt{2} \cos B$ 得 $\cos B$ 的值即可;

(2) 首先求出 A, B, C , 然后由正弦定理可将 a, b 均用含有 c 的式子表示, 结合三角形面积公式即可列方程求解.

【小问 1 详解】

由余弦定理有 $a^2 + b^2 - c^2 = 2ab \cos C$, 对比已知 $a^2 + b^2 - c^2 = \sqrt{2}ab$,

$$\text{可得 } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{\sqrt{2}ab}{2ab} = \frac{\sqrt{2}}{2},$$

因为 $C \in (0, \pi)$, 所以 $\sin C > 0$,

$$\text{从而 } \sin C = \sqrt{1 - \cos^2 C} = \sqrt{1 - \left(\frac{\sqrt{2}}{2}\right)^2} = \frac{\sqrt{2}}{2},$$

又因为 $\sin C = \sqrt{2} \cos B$, 即 $\cos B = \frac{1}{2}$,

注意到 $B \in (0, \pi)$,

所以 $B = \frac{\pi}{3}$.

【小问 2 详解】

由 (1) 可得 $B = \frac{\pi}{3}$, $\cos C = \frac{\sqrt{2}}{2}$, $C \in (0, \pi)$, 从而 $C = \frac{\pi}{4}$, $A = \pi - \frac{\pi}{3} - \frac{\pi}{4} = \frac{5\pi}{12}$,

$$\text{而 } \sin A = \sin \frac{5\pi}{12} = \sin \left(\frac{\pi}{4} + \frac{\pi}{6}\right) = \frac{\sqrt{6} + \sqrt{2}}{4},$$

$$\text{由正弦定理有 } \frac{a}{\sin \frac{5\pi}{12}} = \frac{b}{\sin \frac{\pi}{3}} = \frac{c}{\sin \frac{\pi}{4}},$$

$$\text{从而 } a = \frac{\sqrt{6} + \sqrt{2}}{4} \cdot \sqrt{2}c = \frac{\sqrt{3} + 1}{2}c, b = \frac{\sqrt{3}}{2} \cdot \sqrt{2}c = \frac{\sqrt{6}}{2}c,$$

由三角形面积公式可知, $\triangle ABC$ 的面积可表示为

$$S_{\triangle ABC} = \frac{1}{2}ab \sin C = \frac{3 + \sqrt{3}}{8}c^2,$$

由已知 $\triangle ABC$ 的面积为 $3 + \sqrt{3}$, 可得 $\frac{3 + \sqrt{3}}{8}c^2 = 3 + \sqrt{3}$,

所以 $c = 2\sqrt{2}$.

□

53. (2024 全国 2) 记 $\triangle ABC$ 的内角 A, B, C 的对边分别为 a, b, c , 已知 $\sin A + \sqrt{3} \cos A = 2$.

(1) 求 A .

(2) 若 $a = 2$, $\sqrt{2}b \sin C = c \sin 2B$, 求 $\triangle ABC$ 的周长.

解答 【答案】 (1) $A = \frac{\pi}{6}$

(2) $2 + \sqrt{6} + 3\sqrt{2}$

【解析】

【分析】 (1) 根据辅助角公式对条件 $\sin A + \sqrt{3} \cos A = 2$ 进行化简处理即可求解, 常规方法还可利用同角三角函数的关系解方程组, 亦可利用导数, 向量数量积公式, 万能公式解决;

(2) 先根据正弦定理边角互化算出 B , 然后根据正弦定理算出 b, c 即可得出周长.

【小问 1 详解】

方法一: 常规方法 (辅助角公式)

由 $\sin A + \sqrt{3} \cos A = 2$ 可得 $\frac{1}{2} \sin A + \frac{\sqrt{3}}{2} \cos A = 1$, 即 $\sin(A + \frac{\pi}{3}) = 1$,

由于 $A \in (0, \pi) \Rightarrow A + \frac{\pi}{3} \in (\frac{\pi}{3}, \frac{4\pi}{3})$, 故 $A + \frac{\pi}{3} = \frac{\pi}{2}$, 解得 $A = \frac{\pi}{6}$

方法二: 常规方法 (同角三角函数的基本关系)

由 $\sin A + \sqrt{3} \cos A = 2$, 又 $\sin^2 A + \cos^2 A = 1$, 消去 $\sin A$ 得到:

$4 \cos^2 A - 4\sqrt{3} \cos A + 3 = 0 \Leftrightarrow (2 \cos A - \sqrt{3})^2 = 0$, 解得 $\cos A = \frac{\sqrt{3}}{2}$,

又 $A \in (0, \pi)$, 故 $A = \frac{\pi}{6}$

方法三: 利用极值点求解

设 $f(x) = \sin x + \sqrt{3} \cos x$ ($0 < x < \pi$), 则 $f(x) = 2 \sin(x + \frac{\pi}{3})$ ($0 < x < \pi$),

显然 $x = \frac{\pi}{6}$ 时, $f(x)_{\max} = 2$, 注意到 $f(A) = 2 \sin(A + \frac{\pi}{3}) = 2$,

$f(x)_{\max} = f(A)$, 在开区间 $(0, \pi)$ 上取到最大值, 于是 $x = A$ 必定是极值点,

即 $f'(A) = 0 = \cos A - \sqrt{3} \sin A$, 即 $\tan A = \frac{\sqrt{3}}{3}$,

又 $A \in (0, \pi)$, 故 $A = \frac{\pi}{6}$

方法四: 利用向量数量积公式 (柯西不等式)

设 $a = (1, \sqrt{3}), b = (\sin A, \cos A)$, 由题意, $a \cdot b = \sin A + \sqrt{3} \cos A = 2$,

根据向量的数量积公式, $a \cdot b = |a||b| \cos \langle a, b \rangle = 2 \cos \langle a, b \rangle$,

则 $2 \cos \langle a, b \rangle = 2 \Leftrightarrow \cos \langle a, b \rangle = 1$, 此时 $\langle a, b \rangle = 0$, 即 a, b 同向共线,

根据向量共线条件, $1 \cdot \cos A = \sqrt{3} \cdot \sin A \Leftrightarrow \tan A = \frac{\sqrt{3}}{3}$,

又 $A \in (0, \pi)$, 故 $A = \frac{\pi}{6}$

方法五: 利用万能公式求解

设 $t = \tan \frac{A}{2}$, 根据万能公式, $\sin A + \sqrt{3} \cos A = 2 = \frac{2t}{1+t^2} + \frac{\sqrt{3}(1-t^2)}{1+t^2}$,

整理可得, $t^2 - 2(2 - \sqrt{3})t + (2 - \sqrt{3})^2 = 0 = (t - (2 - \sqrt{3}))^2$,

解得 $\tan \frac{A}{2} = t = 2 - \sqrt{3}$, 根据二倍角公式, $\tan A = \frac{2t}{1-t^2} = \frac{\sqrt{3}}{3}$,

又 $A \in (0, \pi)$, 故 $A = \frac{\pi}{6}$

【小问 2 详解】

由题设条件和正弦定理

$$\sqrt{2}b \sin C = c \sin 2B \Leftrightarrow \sqrt{2} \sin B \sin C = 2 \sin C \sin B \cos B,$$

又 $B, C \in (0, \pi)$, 则 $\sin B \sin C \neq 0$, 进而 $\cos B = \frac{\sqrt{2}}{2}$, 得到 $B = \frac{\pi}{4}$,

于是 $C = \pi - A - B = \frac{7\pi}{12}$,

$$\sin C = \sin(\pi - A - B) = \sin(A + B) = \frac{\sqrt{2} + \sqrt{6}}{4},$$

由正弦定理可得, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, 即 $\frac{2}{\sin \frac{\pi}{6}} = \frac{b}{\sin \frac{\pi}{4}} = \frac{c}{\sin \frac{7\pi}{12}}$,

解得 $b = 2\sqrt{2}, c = \sqrt{6} + \sqrt{2}$,

故 $\triangle ABC$ 的周长为 $2 + \sqrt{6} + 3\sqrt{2}$

□

54. (2024 天津卷) 在 $\triangle ABC$ 中, 角 A, B, C 所对的边分别为 a, b, c , 已知 $\cos B = \frac{9}{16}$, $b = 5$, $\frac{a}{c} = \frac{2}{3}$.

(1) 求 a ;

(2) 求 $\sin A$;

(3) 求 $\cos(B - 2A)$ 的值.

解答 【答案】 (1) 4

(2) $\frac{\sqrt{7}}{4}$

(3) $\frac{57}{64}$

【解析】

【分析】 (1) $a = 2t$, $c = 3t$, 利用余弦定理即可得到方程, 解出即可;

(2) 法一: 求出 $\sin B$, 再利用正弦定理即可; 法二: 利用余弦定理求出 $\cos A$, 则得到 $\sin A$;

(3) 法一: 根据大边对大角确定 A 为锐角, 则得到 $\cos A$, 再利用二倍角公式和两角差的余弦公式即可; 法二: 直接利用二倍角公式和两角差的余弦公式即可.

【小问 1 详解】

设 $a = 2t$, $c = 3t$, $t > 0$, 则根据余弦定理得 $b^2 = a^2 + c^2 - 2ac \cos B$,

即 $25 = 4t^2 + 9t^2 - 2 \times 2t \times 3t \times \frac{9}{16}$, 解得 $t = 2$ (负舍);

则 $a = 4, c = 6$.

【小问 2 详解】

法一: 因为 B 为三角形内角, 所以 $\sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \left(\frac{9}{16}\right)^2} = \frac{5\sqrt{7}}{16}$,

再根据正弦定理得 $\frac{a}{\sin A} = \frac{b}{\sin B}$, 即 $\frac{4}{\sin A} = \frac{5}{\frac{5\sqrt{7}}{16}}$, 解得 $\sin A = \frac{\sqrt{7}}{4}$,

法二: 由余弦定理得 $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 6^2 - 4^2}{2 \cdot 5 \cdot 6} = \frac{3}{4}$,

因为 $A \in (0, \pi)$, 则 $\sin A = \sqrt{1 - \left(\frac{3}{4}\right)^2} = \frac{\sqrt{7}}{4}$

【小问 3 详解】

法一：因为 $\cos B = \frac{9}{16} > 0$ ，且 $B \in (0, \pi)$ ，所以 $B \in (0, \frac{\pi}{2})$ ，

由 (2) 法一知 $\sin B = \frac{5\sqrt{7}}{16}$ ，

因为 $a < b$ ，则 $A < B$ ，所以 $\cos A = \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2} = \frac{3}{4}$ ，

则 $\sin 2A = 2 \sin A \cos A = 2 \times \frac{\sqrt{7}}{4} \times \frac{3}{4} = \frac{3\sqrt{7}}{8}$ ， $\cos 2A = 2 \cos^2 A - 1 = 2 \left(\frac{3}{4}\right)^2 - 1 = \frac{1}{8}$
 $\cos(B - 2A) = \cos B \cos 2A + \sin B \sin 2A = \frac{57}{64}$ 。

法二： $\sin 2A = 2 \sin A \cos A = 2 \times \frac{\sqrt{7}}{4} \times \frac{3}{4} = \frac{3\sqrt{7}}{8}$ ，

则 $\cos 2A = 2 \cos^2 A - 1 = 2 \left(\frac{3}{4}\right)^2 - 1 = \frac{1}{8}$ ，

因为 B 为三角形内角，所以 $\sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \left(\frac{9}{16}\right)^2} = \frac{5\sqrt{7}}{16}$ ，

所以 $\cos(B - 2A) = \cos B \cos 2A + \sin B \sin 2A = \frac{57}{64}$

□

55. (2024 北京卷) 在 $\triangle ABC$ 中，内角 A, B, C 的对边分别为 a, b, c ， $\angle A$ 为钝角， $a = 7$ ， $\sin 2B = \frac{\sqrt{3}}{7} b \cos B$ 。

(1) 求 $\angle A$ ；

(2) 从条件①、条件②、条件③这三个条件中选择一个作为已知，使得 $\triangle ABC$ 存在，求 $\triangle ABC$ 的面积。

条件①： $b = 7$ ；条件②： $\cos B = \frac{13}{14}$ ；条件③： $c \sin A = \frac{5}{2} \sqrt{3}$ 。

注：如果选择的条件不符合要求，第 (2) 问得 0 分；如果选择多个符合要求的条件分别解答，按第一个解答计分。

解答 【答案】(1) $A = \frac{2\pi}{3}$ ；

(2) 选择①无解；选择②和③ $\triangle ABC$ 面积均为 $\frac{15\sqrt{3}}{4}$ 。

【解析】

【分析】(1) 利用正弦定理即可求出答案；

(2) 选择①，利用正弦定理得 $B = \frac{\pi}{3}$ ，结合 (1) 问答案即可排除；选择②，首先求出 $\sin B = \frac{3\sqrt{3}}{14}$ ，再代入式子得 $b = 3$ ，再利用两角和的正弦公式即可求出 $\sin C$ ，最后利用三角形面积公式即可；选择③，首先得到 $c = 5$ ，再利用正弦定理得到 $\sin C = \frac{5\sqrt{3}}{14}$ ，再利用两角和的正弦公式即可求出 $\sin B$ ，最后利用三角形面积公式即可；

【小问 1 详解】

由题意得 $2 \sin B \cos B = \frac{\sqrt{3}}{7} b \cos B$ ，因为 A 为钝角，

则 $\cos B \neq 0$ ，则 $2 \sin B = \frac{\sqrt{3}}{7} b$ ，则 $\frac{b}{\sin B} = \frac{2}{\frac{\sqrt{3}}{7}} = \frac{a}{\sin A} = \frac{7}{\sin A}$ ，解得 $\sin A = \frac{\sqrt{3}}{2}$ ，

因为 A 为钝角，则 $A = \frac{2\pi}{3}$ 。

【小问 2 详解】

选择① $b = 7$, 则 $\sin B = \frac{\sqrt{3}}{14}b = \frac{\sqrt{3}}{14} \cdot 7 = \frac{\sqrt{3}}{2}$, 因为 $A = \frac{2\pi}{3}$, 则 B 为锐角, 则 $B = \frac{\pi}{3}$,

此时 $A + B = \pi$, 不合题意, 舍弃;

选择② $\cos B = \frac{13}{14}$, 因为 B 为三角形内角, 则 $\sin B = \sqrt{1 - \left(\frac{13}{14}\right)^2} = \frac{3\sqrt{3}}{14}$,

则代入 $2\sin B = \frac{\sqrt{3}}{7}b$ 得 $2 \times \frac{3\sqrt{3}}{14} = \frac{\sqrt{3}}{7}b$, 解得 $b = 3$,

$$\sin C = \sin(A + B) = \sin\left(\frac{2\pi}{3} + B\right) = \sin \frac{2\pi}{3} \cos B + \cos \frac{2\pi}{3} \sin B$$

$$= \frac{\sqrt{3}}{2} \times \frac{13}{14} + \left(-\frac{1}{2}\right) \times \frac{3\sqrt{3}}{14} = \frac{5\sqrt{3}}{14},$$

$$\text{则 } S_{\triangle ABC} = \frac{1}{2}ab \sin C = \frac{1}{2} \cdot 7 \cdot 3 \cdot \frac{5\sqrt{3}}{14} = \frac{15\sqrt{3}}{4}.$$

选择③ $c \sin A = \frac{5}{2}\sqrt{3}$, 则有 $c \times \frac{\sqrt{3}}{2} = \frac{5}{2}\sqrt{3}$, 解得 $c = 5$,

则由正弦定理得 $\frac{a}{\sin A} = \frac{c}{\sin C}$, 即 $\frac{7}{\frac{\sqrt{3}}{2}} = \frac{5}{\sin C}$, 解得 $\sin C = \frac{5\sqrt{3}}{14}$,

因为 C 为三角形内角, 则 $\cos C = \sqrt{1 - \left(\frac{5\sqrt{3}}{14}\right)^2} = \frac{11}{14}$,

$$\text{则 } \sin B = \sin(A + C) = \sin\left(\frac{2\pi}{3} + C\right) = \sin \frac{2\pi}{3} \cos C + \cos \frac{2\pi}{3} \sin C$$

$$= \frac{\sqrt{3}}{2} \times \frac{11}{14} + \left(-\frac{1}{2}\right) \times \frac{5\sqrt{3}}{14} = \frac{3\sqrt{3}}{14},$$

$$\text{则 } S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{1}{2} \cdot 7 \cdot 5 \cdot \frac{3\sqrt{3}}{14} = \frac{15\sqrt{3}}{4}$$

□